Journal of Electronic Materials (2026) 55:547–564
https://doi.org/10.1007/s11664-025-12528-9

ORIGINAL RESEARCH ARTICLE

Synergistic Design of Cu-Zn Ferrite–g-C_3N_4 Hybrid Nanocomposite for Solar-Driven Pollutant Degradation and Dye-Sensitized Solar-Cell Applications

Veeranan Arunprasad[1] · Selvam Thulasi[2] · M. M. Abdulrasool[3] · M. Hema Kumar[4]

Received: 18 June 2025 / Accepted: 25 October 2025 / Published online: 12 November 2025
© The Minerals, Metals & Materials Society 2025

Abstract

The simultaneous demand for clean energy and water purification drives the development of multifunctional photocatalysts and electrocatalysts. We sought to fabricate a $Cu_{0.5}$ $Zn_{0.5}$ Fe_2O_4–graphitic carbon nitride hybrid using ultrasonic-assisted hydrothermal processing and assess its efficacy for sun-driven degradation of rhodamine B and as counter electrodes in dye-sensitized solar cells (DSSCs). X-ray Diffraction (XRD) validates the formation of pure spinel $Cu_{0.5}$ $Zn_{0.5}$ Fe_2O_4 phase and the presence of g-C_3N_4. X-ray photoelectron spectroscopy (XPS) study demonstrates the interfacial electronic interaction. Field emission scanning electron microscopy (FESEM) and transmission electron microscopy (TEM) results demonstrate that $Cu_{0.5}$ $Zn_{0.5}$ Fe_2O_4 nanoparticles are uniformly affixed to g-C_3N_4 sheets. Brunauer–Emmett–Teller (BET) analysis indicates a mesoporous structure with a surface area of 33.7 m^2 g^{-1}. UV–Vis diffuse reflectance spectroscopy (DRS) results yield a refined E_g of 2.20 eV and photoluminescence (PL) quenching signifies effective charge separation. Under natural sunlight, the optimized $Cu_{0.5}$ $Zn_{0.5}$ Fe_2O_4–g-C_3N_4(25%) degrades 94.6% of RhB in 90 min. The dye-sensitized solar cell (DSSC) counter electrode has a power conversion efficiency of 6.14% (J_{sc} = 15.06 mA cm^{-2}, V_{oc} = 0.75 V, FF = 0.64), with a charge transfer resistance (R_{ct}) of 6.25 Ω, a high saturation current density (J_0) of 0.92 mA cm^{-2}, and almost 95% power conversion efficiency (PCE) retention after 15 days. This study presents, for the first time, a CZFO–g-C_3N_4 heterojunction that concurrently serves as a visible-light photocatalyst and a platinum-free counter electrode for dye-sensitized solar cells, providing a sustainable approach that integrates pollutant degradation with solar energy conversion.

✉ Veeranan Arunprasad
arunprasad2978@gmail.com

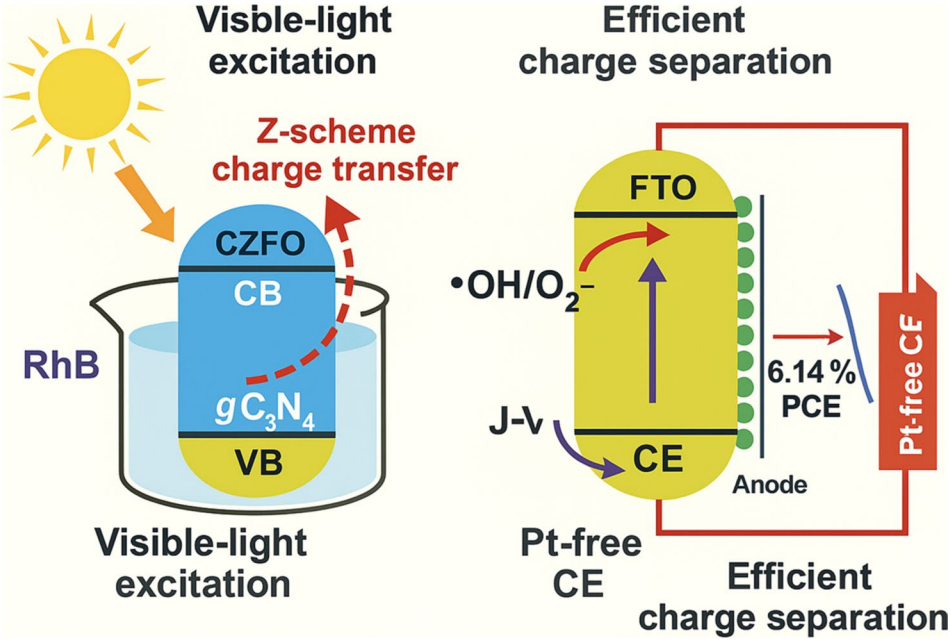
1 Department of Mechanical Engineering, Theni Kammavar Sangam College of Technology, Koduvilarpatti, Theni, Tamil Nadu 625534, India

2 Department of Mechanical Engineering, University College of Engineering, BIT Campus, Anna University, Tiruchirappalli, Tamil Nadu 620024, India

3 Department of Chemistry, College of Science, University of Kerbala, Kerbala, Iraq

4 Department of Electronics and Communication Engineering, Sona College of Technology, Salem, Tamil Nadu 636005, India

Graphical Abstract



Keywords $Cu_{0.5}Zn_{0.5}Fe_2O_4$–g-C_3N_4 hybrid · counter electrode · rhodamine B degradation · interfacial charge transfer · visible-light activity

Introduction

In recent decades, rapid industrialization, population growth, and the extensive discharge of chemical effluents have led to a severe global water-pollution crisis. Hazardous dyes, pharmaceutical waste, and industrial chemical effluents are major contributors to water pollution. These contaminants are resistant to degradation and can significantly threaten environmental integrity and human health.[1] Persistent contaminants, such as synthetic dyes, pharmaceutical residues, and heavy metals, are challenging to breakdown using conventional methods such as adsorption or biological treatment, which only relocate pollutants from one phase to another rather than eliminating them.[2] In this context, semiconductor-based photocatalytic degradation has emerged as a viable, ecofriendly, and economical method for the complete mineralization of hazardous organics into innocuous by-products such as CO_2 and H_2O under solar or visible light irradiation.[3] Numerous studies using TiO_2, ZnO, WO_3, and Fe_2O_3 have demonstrated the effectiveness of semiconductor photocatalysts in wastewater treatment.[4] However, these materials still suffer from narrow light absorption ranges and fast electron–hole recombination, limiting large-scale applicability. At the same time, the increasing global energy demand and the depletion of fossil resources have heightened the quest for clean, renewable energy technologies.[5] Dye-sensitized solar cells (DSSCs) have garnered interest owing to their affordability, ease of fabrication, and exceptional performance in low-light environments.[6] Photocatalytic water treatment and DSSCs both operate on the fundamental principle of light-induced charge production and transfer at semiconductor surfaces. Therefore, the development of multifunctional materials capable of concurrently harvesting sunlight for pollutant degradation and energy conversion presents a sustainable dual-purpose solution.

In this regard, researchers have recently focused much of their interest on ferrite-based materials (general formula MFe_2O_4) owing to their magnetic nature, tunable bandgap, structural stability, and excellent redox properties.[7,8] These features make them highly suitable for a variety of applications, including photocatalysis, electrocatalysis, sensors, super capacitors, and solar energy conversion.[9] The presence of both divalent and trivalent cations in the spinel lattice promotes efficient charge transport and surface catalytic activity, which are crucial for both photo degradation and photovoltaic processes. Particularly mixed ferrites, such as Cu-Zn, Ni-Co, and Mn-Fe-based spinels, have drawn a lot of interest because of their improved physicochemical characteristics over their single-metal equivalents.[10–12] Two distinct divalent metal ions incorporated into the ferrite lattice causes lattice distortions, change the bandgap, increases surface active sites and enhances the mobility of charge carriers, therefore

improving their performance in photocatalytic and photovoltaic uses. For pollution degradation under visible light as well as for energy conversion systems, Cu-Zn ferrites ($Cu_xZn_{1-x}Fe_2O_4$) show a special mix of optical absorption, redox activity, and magnetic separability.[10] Nevertheless, ferrites continue to experience limited visible-light absorption and rapid electron–hole recombination. Constructing a heterostructure by coupling them with a second semiconductor is a practicable solution that enhances charge separation and broadens light harvesting. In this context, in order to construct Z-scheme composites with separability, graphitic carbon nitride (g-C_3N_4) has been chosen as a co-catalyst owing to its exceptional band structure, stability, and responsiveness to visible light.[13] This makes it an ideal choice for the degradation of pollutants under visible light and for energy conversion systems. However, g-C_3N_4's quick recombination of photogenerated electron–hole pairs and weak electrical conductivity frequently restrict it when used alone.[14] As a result, combining it with mixed ferrites, such Cu-Zn ferrite, creates a type-II or Z-scheme heterojunction that may greatly enhance interfacial charge separation and increase absorption of visible light. In addition to increasing photocatalytic efficiency for pollutant degradation, this synergistic hybridization also increases redox kinetics and interfacial electron transport, both of which are advantageous for DSSC performance.[15] The effectiveness of such hybrid systems has been shown in recent research. In one study, a reduced graphene oxide (rGO) loaded with $Co_{0.7}Zn_{0.3}Er_xFe_{2-x}O_4$ magnetic nanoparticles was successfully synthesized.[16] Under visible light irradiation, the rGO demonstrated enhanced photocatalytic degradation of methylene blue dye, reaching a complete degradation efficiency after 1 h. This enhancement was ascribed to the carbon-based materials and ferrites synergistic actions, which promoted prolonged light absorption and effective charge separation. Similarly, the addition of g-C_3N_4 to photoanode materials has demonstrated encouraging outcomes in the field of DSSCs. According to research, a heterostructure g-C_3N_4–$CuFe_2O_4$ nanocomposite that was employed as a photoanode in solid-state DSSCs outperformed the pristine $CuFe_2O_4$-based cells with a power conversion efficiency (PCE) of 7.92%.[17] The increased light absorption and charge transfer made possible by the g-C_3N_4 and ferrite components were credited with the improvement. The microstructural and interfacial characteristics of thin films are crucial in influencing the performance of DSSCs. Factors including film porosity, surface roughness, and interparticle connection directly affect dye loading, light scattering, and electron transport inside the photoelectrode or counter-electrode layers.[18] Optimally designed thin films enhance the interaction between the active material and the underlying FTO substrate, promoting efficient charge collection and reducing recombination losses.[19] Furthermore, in counter-electrodes, consistent film shape and numerous catalytic active sites enhance the triiodide reduction reaction and reduce charge-transfer resistance, both of which are critical for attaining high solar efficiency.[20] The creation of uniform, porous, and conductive thin-film structures with sustainable materials including ferrite–carbon nitride hybrids present a viable alternative to expensive platinum electrodes. Recent investigations indicate that optimized ferrite-based or carbon-modified thin films can attain power conversion efficiencies comparable to noble-metal systems, while maintaining exceptional stability and cost-effectiveness.[21]

Based on the above considerations, the selection of $Cu_{0.5}Zn_{0.5}Fe_2O_4$ arises from the synergistic functions of Cu^{2+} and Zn^{2+} ions within the spinel lattice. Zn^{2+} ions maintain the structure via tetrahedral occupancy, whereas Cu^{2+} ions at octahedral sites promote Fe^{3+}–Fe^{2+} redox transitions and improve conductivity. The lattice attains balanced structural stability and redox activity in an equimolar 0.5:0.5 ratio, resulting in an optimum visible-light bandgap of around 2.3–2.5 eV. The integration of $Cu_{0.5}Zn_{0.5}Fe_2O_4$ with g-C_3N_4 alleviates the inherent limitations of each constituent. Although g-C_3N_4 exhibits robust visible-light absorption and a conjugated π-network, it is plagued by fast electron–hole recombination. The integration of $Cu_{0.5}Zn_{0.5}Fe_2O_4$ establishes a Z-scheme (type-II-like) heterojunction, facilitating the transfer of electrons from the conduction band of g-C_3N_4 (about −1.13 eV) to that of $Cu_{0.5}Zn_{0.5}Fe_2O_4$ (approximately −0.53 eV), while holes are retained in the valence band of g-C_3N_4 (approximately +1.57 eV). This interfacial charge transfer inhibits recombination and facilitates redox processes for both dye degradation and triiodide reduction. This study seeks to produce $Cu_{0.5}Zn_{0.5}Fe_2O_4$–g-C_3N_4 hybrid nanocomposites using an ultrasonic-assisted hydrothermal technique, assess their photocatalytic efficacy under visible light, and investigate their potential as counter-electrode materials in DSSCs. The logical band-gap alignment and heterojunction design position CZFO–CN as a sustainable, multifunctional material for concurrent wastewater purification and solar energy conversion.

Experimental Procedure

Materials

Copper(II) nitrate trihydrate ($Cu(NO_3)_2{\cdot}3H_2O$, > 99%), zinc nitrate hexahydrate ($Zn(NO_3)_2{\cdot}6H_2O$, > 98%), ferric nitrate nonahydrate ($Fe(NO_3)_3{\cdot}9H_2O$, > 98%), citric acid ($C_6H_8O_7$, > 99.5%), melamine ($C_3H_6N_6$, > 99%), N 719 dye, Rhodamine B (RhB dye, > 95%) were purchased from SRL Chemicals, Chennai, India, and used without further purification. Deionized water was employed throughout the experimental work.

Synthesis of g-C_3N_4, $Cu_{0.5}Zn_{0.5}Fe_2O_4$ and $Cu_{0.5}Zn_{0.5}Fe_2O_4$–g-C_3N_4 Composites

Graphitic carbon nitride (g-C_3N_4) was synthesized via a simple thermal polycondensation method.[22] In detail, 10 g of melamine powder was placed in a covered ceramic crucible, heated up to 500°C with a heating rate of 10°C/min, and maintained at 500°C for 4 h. The product was collected and pulverized into a powder in an agate mortar for subsequent use. To synthesize $Cu_{0.5}$ $Zn_{0.5}$ Fe_2O_4 nanoparticles, 0.120 g of copper nitrate trihydrate, 0.149 g of zinc nitrate hexahydrate, and 0.808 g of ferric nitrate nonahydrate were dissolved in 40 mL of deionized water under constant magnetic stirring. The pH of the solution was adjusted to ~9 using 1 M NaOH solution under slow addition. The homogeneous solution was then subjected to ultrasonic irradiation (40 kHz, 200 W) for 30 min. The sonicated mixture was subsequently transferred to a 100 mL Teflon-lined stainless-steel autoclave and heated at 180°C for 12 h. After natural cooling, the obtained precipitate was collected, washed several times with deionized water and ethanol, and dried at 80°C overnight. The dried powder was then mildly calcined at 400–500°C for 2 h, yielding pure $Cu_{0.5}$ $Zn_{0.5}$ Fe_2O_4 which is labelled as CZFO. To prepare the CZFO–g-C_3N_4 composites, appropriate amounts of CZFO (15 wt.%, 20 wt.%, and 25 wt.%) were dispersed with g-C_3N_4 in ethanol, ultrasonicated for 1 h, and stirred for 2 h. The resulting mixtures were dried at 80°C overnight, ground into fine powders, and labeled as CZFO–CN15, CZFO–CN20, and CZFO–CN25, respectively.

Fabrication of CZFO-CN-Based Counter Electrodes and DSSCs

Fluorine-doped tin oxide (FTO) coated glass substrates (Techinstro, India) were used for device fabrication. The substrates had a sheet resistance of approximately 15 Ω sq^{-1}, a resistivity of 8×10^{-4} Ω cm, and a total surface area of 2.5×2.5 cm^2. The FTO coating thickness was around 500 ± 20 nm, and the transmittance in the visible region was about 80–85%. These specifications were chosen to ensure optimum optical transparency and electrical conductivity for efficient photovoltaic operation. The FTO glass substrates were initially subjected to sequential ultra-sonication in deionized water, acetone, and isopropanol (IPA), each for duration of 15 min. Upon drying, the CZFO–CN composite slurry was formulated by combining 50 mg of the corresponding CZFO–CN15, CZFO–CN20, and CZFO–CN25 powder with few drops of ethanol and 5 µL of Nafion solution as a binder. The resultant slurry was drop-coated onto the cleaned FTO substrates and dried at 80°C for 30 min, thereafter undergoing annealing at 120°C for 1 h. The coated substrates were used as counter electrodes (CEs) to fabricate DSSCs. A conventional Pt CE was fabricated by drop-casting 5 mM H_2PtCl_6 in isopropanol onto a cleaned FTO glass, followed by annealing at 400°C for 30 min. The TiO_2 photoanode was fabricated by doctor-blading a TiO_2 paste onto cleaned FTO glass, thereafter undergoing incremental sintering to 500°C.[23] The sintered TiO_2 photoanode was immersed in a 0.5 mM ethanolic solution of N719 ruthenium dye (cis-diisothiocyanato-bis(2,2′-bipyridyl-4,4′-dicarboxylato)-ruthenium (II)) for 12 h at room temperature to ensure complete dye adsorption, followed by gentle rinsing with ethanol to remove unbound molecules and subsequent drying prior to device assembly. The photoanode and CE were then assembled face-to-face with an iodide/triiodide electrolyte sandwiched between them. The active area of the assembled DSSC was maintained at 0.25 cm^2 using a masking template for accurate photovoltaic measurements. A schematic illustration of the fabrication process shown in Fig. 1a.

Photocatalytic Experiment

Typically, 50 mg of the photocatalyst was dissolved in 100 mL of RhB solution (10 mg/L), with dil. HCl used to bring the original pH down to about 6.5. In order to achieve adsorption–desorption equilibrium, the suspension was agitated for 30 min in the dark. The reaction mixture was constantly agitated during the irradiation process, while the surrounding temperature fluctuated between 32°C and 36°C. A UV–Vis spectrophotometer was used to measure the absorbance of the supernatant at 554 nm after 5 mL aliquots were taken out at regular intervals of 15 min and centrifuged at 6000 rpm to eliminate particles. To verify that the deterioration was photocatalytic, control studies were conducted without light or photocatalyst. Throughout the experiment, the reactor's effective irradiation area was kept at around 25 cm^2. The photocatalytic efficiency was calculated using the standard equation[24]:

$$Degradation(\%) = \frac{C_0 - C_t}{C_0} \times 100 \quad (1)$$

where C_0 and C_t are initial concentration of dye (mg/L) and concentration at time *t* respectively.

Characterization

The crystalline structure of the synthesized samples was examined using an x-ray diffractometer (Rigaku Miniflex 600, Japan) operating with Cu K_α radiation ($\lambda = 1.5406$ Å). The patterns were recorded in the 2θ range of 10–80°, with a step size of 0.02°, scan rate of 2° min^{-1}, and a Ni filter to eliminate K_β radiation. The accelerating voltage and current were maintained at 40 kV and 15 mA, respectively. The KBr pellet technique was employed to record Fourier-transform infrared (FTIR) spectra

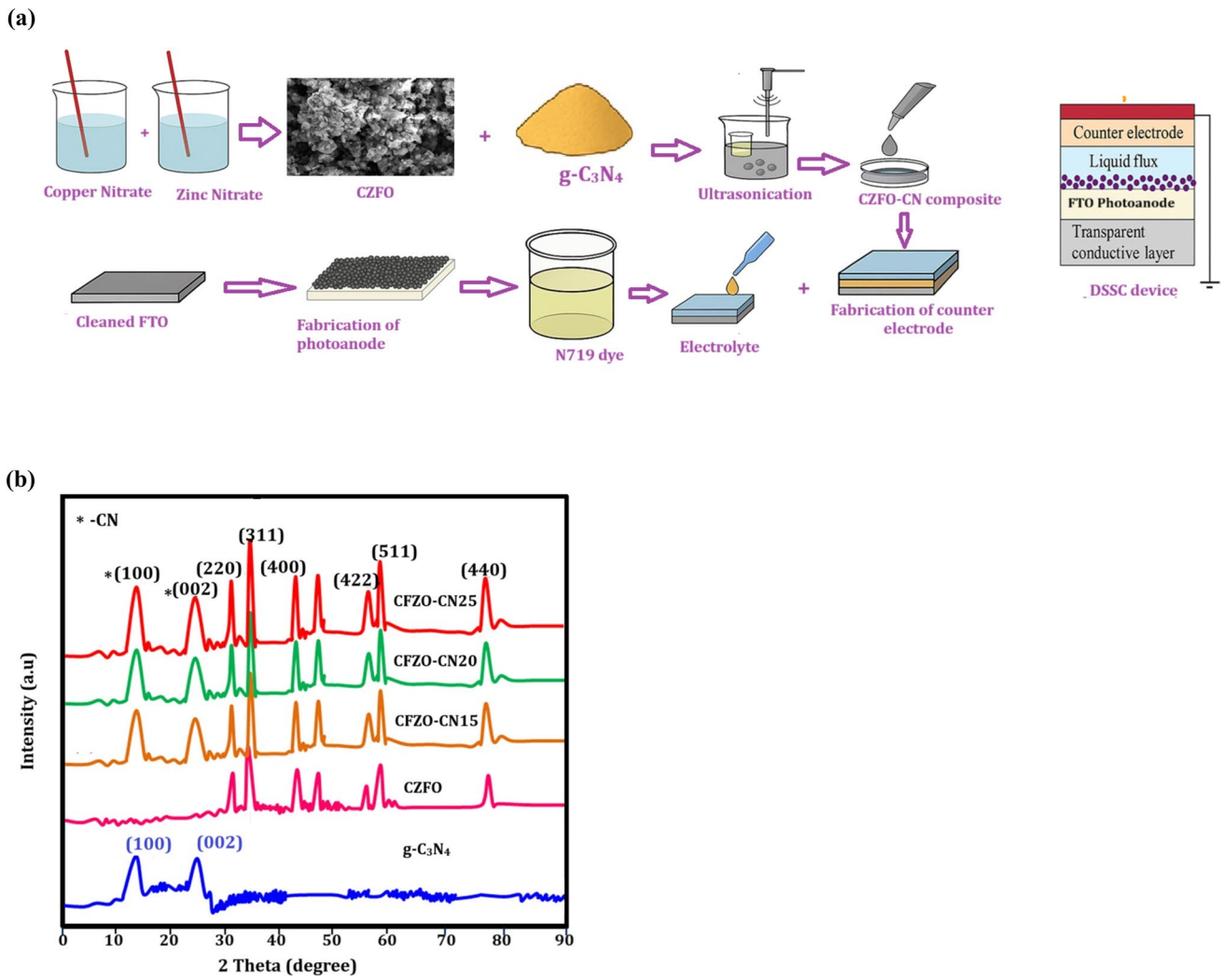


Fig. 1 (a) Schematic illustration of fabrication of dye-sensitized solar cells (DSSC) (b) x-ray diffraction (XRD) patterns of g-C_3N_4, CZFO, and CZFO–CN composites.

in the 400–4000 cm^{-1} range using a Bruker Alpha spectrometer with a resolution of 4 cm^{-1}. A field-emission scanning electron microscope (FESEM, FEI Quanta FEG 250) operating at an accelerating voltage of 10–15 kV was used to study the surface morphology, and a transmission electron microscope (TEM, JEOL JEM-2100) operating at 200 kV was used to do a comprehensive microstructural investigation. The Micromeritics ASAP 2020 analyzer was employed to determine the specific surface area and pore structure at 77 K, utilizing N_2 adsorption–desorption isotherms. The Brunauer–Emmett–Teller (BET) and Barrett–Joyner–Halenda (BJH) procedures were employed to estimate the surface area, pore volume, and pore diameter of the samples, which were degassed at 150°C for 6 h before analysis. Photoluminescence (PL) spectra were obtained using a Horiba Jobin Yvon Fluorolog-3 spectrofluorometer with an excitation wavelength of 325 nm, a slit width of 2 nm, and a scan speed of 120 nm min^{-1}. We used a Thermo Scientific K-Alpha spectrometer with Al K_α radiation (hv = 1486.6 eV) and an ultra-high vacuum (1.33×10^{-10} kPa) to do x-ray photoelectron spectroscopy (XPS) studies. UV–Visible diffuse reflectance spectra (UV–Vis DRS) were collected using a Shimadzu UV-2600 spectrophotometer in the wavelength range of 200–800 nm. Photovoltaic measurements were performed using a Newport Oriel solar simulator (AM 1.5G, 100 mW cm^{-2}) connected to a Keithley 2400 source meter, and the photocatalytic degradation studies were monitored by a Shimadzu UV-2600 spectrophotometer.

Results and Discussion

Powder XRD Analysis

The crystalline phases and structural purity of the synthesized samples were investigated by XRD patterns and the results are depicted in Fig. 1b. The XRD pattern of pristine CZFO shows clear peaks at 30.2°, 35.4°, 43.2°, 53.5°, 57.0°, and 62.6°. These peaks correspond to the (220), (311), (400), (422), (511), and (440) planes. The patterns match well with the cubic spinel structure of $Cu_{0.5}$ $Zn_{0.5}$ Fe_2O_4 (JCPDS No. 77-0010).[25] The XRD pattern of pristine g-C_3N_4 displays two prominent diffraction peaks at roughly 13.1° and 27.4°, corresponding to the (100) and (002) planes, respectively. These diffraction peaks are in good agreement with the standard JCPDS card No. 87-1526, corresponding to the (100) and (002) planes of graphitic carbon nitride, thereby confirming the successful formation of polymeric g-C_3N_4.[26] The peaks correspond to the in-plane structural arrangement of tri-s-triazine units and the interlayer stacking of conjugated aromatic layers. In this instance, both peaks exhibit roughly equivalent intensity, suggesting partial restacking and diminished layer ordering in the g-C_3N_4 framework, which often occurs during hot polymerization. Similar intensity variations have been reported in other g-C_3N_4 systems synthesized under comparable conditions.[27] The XRD patterns of the CZFO–CN composites (CZFO–CN15, CZFO–CN20, and CZFO–CN25) exhibit the characteristic diffraction peaks of spinel-type $Cu_{0.5}$ $Zn_{0.5}$ Fe_2O_4 and the (002) reflection of g-C_3N_4 near 27.4°, confirming the coexistence of both phases without the formation of any new crystalline structure. The absence of new or shifted peaks indicates that the composite formation occurs through physical coupling and interfacial interaction rather than chemical substitution or lattice transformation. Moreover, g-C_3N_4 is known for its low crystallinity and weak diffraction intensity due to the presence of light elements (C and N), so its signals are often dominated by the strong spinel peaks of CZFO. The minor variation in peak intensity and slight broadening observed in the composites can be attributed to nanoparticle dispersion, interfacial strain, and confinement effects of CZFO within the g-C_3N_4 matrix.[28] These structural effects do not alter the long-range crystal order but confirm the formation of an intimate heterojunction interface between CZFO and g-C_3N_4. The average crystallite sizes of the CZFO–CN composites, estimated from the most prominent (311) reflection via the Debye–Scherrer equation, were approximately 46 ± 2 nm for CZFO–CN15, 42 ± 2 nm for CZFO–CN20, and 35 ± 2 nm for CZFO–CN25.[29] The gradual decrease in crystallite size with elevated g-C_3N_4 concentration can be ascribed to the growth-confinement and surface-passivation effects of the g-C_3N_4 matrix. In composite creation, the g-C_3N_4 nanosheets serve as a physical barrier, inhibiting the coalescence of CZFO nanoparticles and restricting their grain development. Furthermore, the many functional groups and surface imperfections on g-C_3N_4 act as heterogeneous nucleation sites, facilitating the development of smaller, uniformly distributed CZFO crystallites. The reduction in crystallite size arises from the synergistic interaction of surface confinement and interfacial stabilization, resulting in a uniform distribution of CZFO nanoparticles inside the g-C_3N_4 framework and an overall improvement in the structural homogeneity of the composites.

SEM and TEM Analysis

The morphologies of CZFO, g-C3N4, CZFO15, CZFO20, and CZFO25 are characterized by FESEM and the results

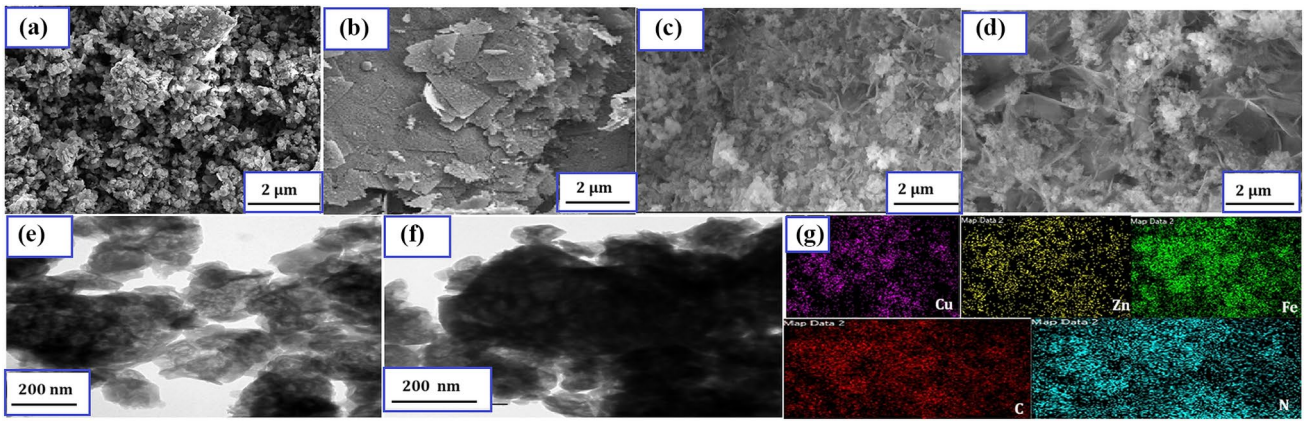


Fig. 2 Scanning electron microscope (SEM) images (a) CZFO (b) g-C_3N_4 (c) CZFO–CN15 (d) CZFO–CN25; transmission electron microscope (TEM) images of (e) CZFO and (f) CZFO–CN25; (g) Elemental mapping of CZFO–CN25.

are depicted in Fig. 2. As shown in Fig. 2a, the pristine CZFO sample consists of irregularly shaped nanoparticles with moderate agglomeration, a feature commonly observed in spinel ferrites synthesized via hydrothermal routes. The particles exhibit a quasi-spherical to polygonal morphology, suggesting nonuniform grain growth and partial aggregation, potentially owing to the elevated surface energy of nanoscale ferrite domains. The pure g-C_3N_4 (Fig. 2b) has plate-like aggregates that are irregularly shaped and have a rough surface. These are caused by the stacking of condensed layers occurring on a micrometer scale when melamine is heated and polymerized. They look as if they are tightly packed and fused unevenly, which is more similar to bulk g-C_3N_4 structures than exfoliated nanosheets. This shape results from some moisture and restacking occurring during the calcination process. The CZFO–CN15 composite samples (Fig. 2c) exhibited partial embedding of CZFO nanoparticles over the g-C_3N_4 sheets, suggesting a high level of physical interaction and dispersion. A denser distribution of CZFO nanoparticles was observed across the g-C_3N_4 surface as the CZFO content increased in CZFO–CN25 (Fig. 2d). This suggests that the interfacial contact and anchoring of ferrite particles onto the carbon nitride matrix were enhanced. Figure 2e shows the TEM image of pristine CZFO, revealing quasi-spherical nanoparticles with an average diameter of 20–30 nm. Particle sizes were quantified using ImageJ (v1.53t) by measuring more than 100 particles. The particles are moderately aggregated, consistent with the FESEM result. Distinct lattice fringes with a spacing of 0.25 nm correspond to the (311) plane of the spinel phase, hence affirming excellent crystallinity. The TEM image (Fig. 2f) depicts fine CZFO nanoparticles uniformly affixed to the g-C_3N_4 sheets, establishing a close solid–solid interface devoid of discernible voids or isolated domains. This close proximity facilitates direct electron transfer between the two components. The attachment primarily results from electrostatic attraction between positively charged CZFO surfaces and negatively polarized nitrogen-rich spots on g-C_3N_4, facilitated by hydrogen-bond interactions during ultrasonication. This physical anchoring ensures the formation of an interfacial heterojunction that enables effective charge separation during photoexcitation. Moreover, the uniform distribution and reduced aggregation relative to virgin CZFO suggest effective growth confinement by the g-C_3N_4 matrix. Elemental mapping (Fig. 2g) verifies the uniform distribution of Cu, Zn, Fe, C, and N, whereas energy-dispersive x-ray spectroscopy (EDX) quantitative analysis provides approximate atomic ratios of Cu (9.6%), Zn (10.2%), Fe (19.8%), C (33.5%), and N (26.9%), confirming the homogeneous composition of the composite.

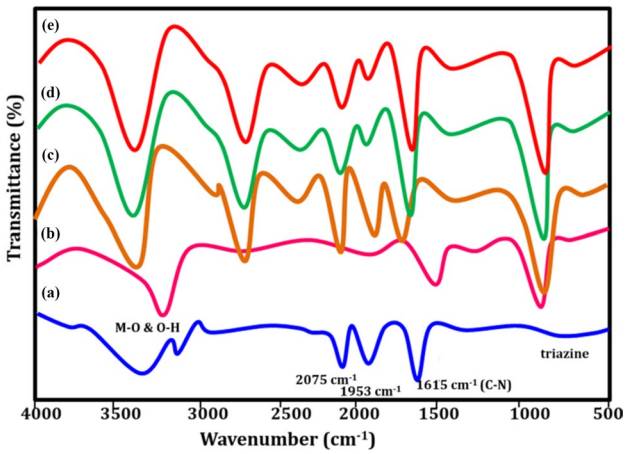


Fig. 3 Fourier transform infrared (FTIR) spectra of (a) g-C_3N_4 (b) CZFO (c) CZFO–CN15 (d) CZFO–CN20 (e) CZFO–CN25.

FTIR Analysis

The FTIR spectra of g-C_3N_4, CZFO, and CZFO–CN composites were recorded in the range of 400–4000 cm^{-1} to identify the functional groups and confirm composite formation. Fig. 3a, b, c, d, and e displays the FTIR spectra of the g-C_3N_4, CZFO, and CZFO–CN composites. As shown in Fig. 3a, three noticeable peaks in g-C_3N_4 can be seen at 1615 cm^{-1}, 1953 cm^{-1}, and 2075 cm^{-1}. The C=N stretching vibration of the heterocyclic aromatic rings is represented by the band at 1615 cm^{-1}. Furthermore, the FTIR spectra of the CZFO–CN composites and g-C_3N_4 exhibit two small bands, which are centered at 2075 cm^{-1} and 1953 cm^{-1}, respectively. These bands are typically ascribed to overtone and combination vibrations that are associated with structural defects and the extended π-conjugated system of the heptazine units in polymeric carbon nitride. These features are typical of thermally condensed g-C_3N_4 and suggest the presence of partially disordered graphitic domains, which is consistent with previous report.[30] In the prepared g-C_3N_4, the absence of a distinct peak at 810 cm^{-1}, which typically corresponds to the triazine ring breathing mode, suggests the possibility of structural disorder or stacking variation. The FTIR spectrum of pristine CZFO exhibits prominent absorption bands at 8480 cm^{-1}, 1050 cm^{-1}, 1271 cm^{-1}, and 3210 cm^{-1}. These bands are related to metal–oxygen (M-O) bond vibrations, lattice distortions, and surface hydroxyl groups.[31] The lattice distortion inferred from these M-O vibrations align with the peak broadening observed in the XRD patterns, particularly near the (311) and (440) planes of the CZFO–CN composites. This correlation confirms that the interaction between

CZFO nanoparticles and the g-C_3N_4 matrix induces minor structural strain at the nanoscale, consistent with both XRD and FTIR analyses. In addition, a broad absorption

band in the 1700–2000 cm^{-1} range is observed, which may be attributed to surface-adsorbed species, oxygen vacancies, or lattice-bound water. These phenomena are prevalent in hydrothermally prepared spinel ferrites.[32] As shown Fig. 3c, d, and e, the spectra of CZFO–CN15, CZFO–CN20, and CZFO–CN25 exhibit characteristics of both g-C_3N_4 and CZFO, therefore validating the successful synthesis of the hybrid. As the CZFO concentration increases, the M-O stretching bands intensify and minor changes in the C-N/C=N area indicate interfacial interactions between CZFO nanoparticles and g-C_3N_4 sheets. Furthermore, in the composites, a strong peak is also discernible at ~2700 cm^{-1}. This feature is attributed to a combination/overtone vibration of g-C_3N_4 N-H modes (approximately twice the ~1350 cm^{-1} bending) with ring vibrations, arising via Fermi resonance. Its clearer appearance in CZFO–CN relative to pristine g-C_3N_4 likely reflects interfacial H-bonding (M-O-H···N/C) and local polarization at the CZFO/g-C_3N_4 junction. These spectral observations, together with the unchanged XRD pattern, confirm that the formation of CZFO–CN composites occur through interfacial coupling rather than the creation of a new crystalline phase. Such vibrational coupling further verifies the strong bonding between the two components at the nanoscale, which contributes to efficient charge transfer and improved catalytic properties.

XPS Analysis

The XPS analysis was performed to examine the surface elemental composition and chemical states of the elements in the CZFO–CN25 composite. The successful incorporation of CZFO and g-C_3N_4 components in the composite is confirmed by the survey spectrum, which validates the presence of Cu, Zn, Fe, C, N, and O (Fig. 4a). The high-resolution Cu 2*p* spectrum has two prominent peaks at 934.2 eV and 954.1 eV, corresponding to Cu $2p_{3/2}$ and Cu $2p_{1/2}$ respectively, along with satellite peaks. This confirms the occurrence of the Cu^{2+} oxidation state in the CZFO lattice.[33] The distinct satellite structure is indicative of Cu^{2+} species inside a ferrite matrix (Fig. 4b). The Zn 2*p* region displays peaks at 1021.8 eV (Zn $2p_{3/2}$) and 1044.9 eV (Zn $2p_{1/2}$), confirming the presence of Zn^{2+} ions. The sharp, well-defined peaks are consistent with zinc in a spinel ferrite configuration.[34] As shown in Fig. 4d, the high-resolution Fe 2*p* spectrum displays two principal peaks at 709.3 eV (Fe $2p_{3/2}$) and 724.6 eV (Fe $2p_{1/2}$), along with distinctive satellite characteristics. This indicates the presence of Fe^{3+} and Fe^{2+} oxidation states, affirming the characteristic mixed-valence nature of Fe in spinel ferrite.[35] Figure 4e displays the deconvoluted XPS data of C 1*s*. The C-C/C=C, C-N and N-C=N bonds are attributed to the three deconvoluted peaks at 285.6 eV, 286.7 eV, and 288.2 eV, respectively. These peaks confirm that the composite contains the conjugated aromatic framework of g-C_3N_4.[36] The fitted curves and XPS findings for N 1*s* are displayed in Fig. 4f. With binding energies of 398.2 eV, 399.4 eV, and 400.4 eV, each fitted curve may be assigned to protonated amino groups (N-H^+), tertiary nitrogen (N-$(C)_3$), or sp^2-hybridized nitrogen (C=N-C).[37] The O 1*s* spectrum exhibits a broad peak deconvoluted into two components at 530.2 eV and 532.2 eV, attributed to lattice

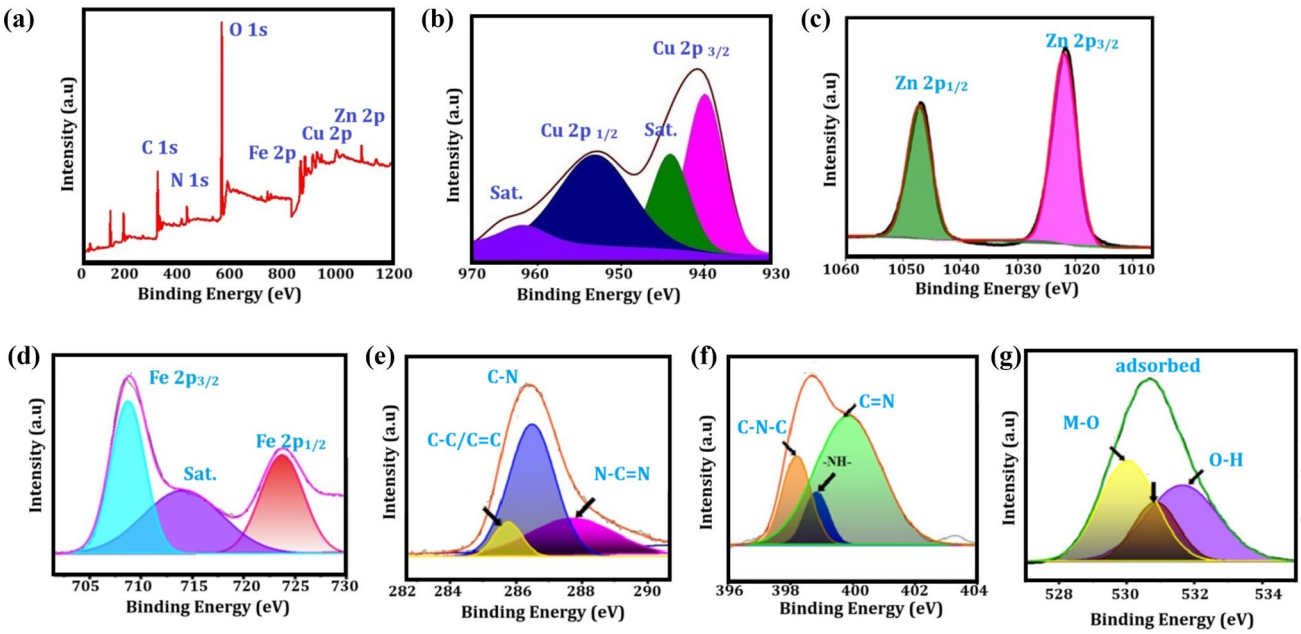


Fig. 4 X-ray photoelectron spectroscopy (XPS) spectra of CZFO–CN25 (a) survey scan (b) Cu 2p (c) Zn 2p (d) Fe 2p (e) C 1s (f) N1s (g) O 1s.

oxygen (M-O) and surface-adsorbed oxygen, or hydroxyl groups, respectively. This signifies the existence of both structural and surface oxygen species inside the composite.[38] Moreover, the quantitative surface composition obtained from the XPS survey spectrum of the CZFO–CN25 composite reveals atomic percentages of Cu (7.9 at%), Zn (8.4 at%), Fe (16.2 at%), C (37.8 at%), N (24.7 at%), and O (5.0 at%), which are in close agreement with the theoretical stoichiometry of the hybrid structure. The presence of all constituent elements and their near-stoichiometric ratios confirm the successful formation of the CZFO/g-C_3N_4 heterostructure and its homogeneous elemental distribution across the surface. Although XRD does not reveal any new diffraction peaks or shifts, XPS analysis confirms electronic communication between the components. The modest broadening of the C 1*s* (C-N/C=N) and N 1*s* (sp^2 N, tertiary N) peaks, in conjunction with the coexistence of Cu, Zn, Fe, C, N, and O signals in a single composite spectrum, implies local charge redistribution across the CZFO/g-C_3N_4 interface. This interfacial polarization is a characteristic of type-II heterojunctions and confirms that the two phases are electronically coupled while maintaining their crystal individuality.

BET Analysis

The specific surface area and porosity of the synthesized samples were evaluated using N_2 adsorption–desorption and BET analysis. The N_2 adsorption-desorption isotherms of CZFO, CZFO-CN15, and CZFO-CN25 are presented in Fig. 5a. All the samples showed a typical type-IV isotherm curve with an H_3-type hysteresis loop at high pressure. According to IUPAC classification, this indicates the presence of mesoporous structures.[39] These are usually formed by layered or slit-like pores. The BET surface area of pure CZFO was calculated at 22.6 m^2/g, which increased to 28.4 m^2/g for CZFO–CN15 and 33.7 m^2/g for CZFO–CN25. This indicates an estimated 49% increase in surface area with the augmentation of g-C_3N_4 concentration. The augmentation is ascribed to the efficient distribution of CZFO nanoparticles inside the g-C_3N_4 matrix, which inhibits restacking and agglomeration, thus enhancing the quantity of accessible active sites.[40] A steady increase was noted in total pore volume (from 0.147 cc/g to 0.213 cc/g) and average pore diameter (from 15.8 nm to 18.6 nm), signifying an enhanced mesoporous architecture. The values are consolidated in Table I. These findings are further corroborated by the BJH pore size distribution plots displayed in Fig. 5b. The plot shows that the majority of the samples have pores with diameters in the 15-20 nm mesoporous range. In the case of composites, it could be seen somewhat wider distribution. This is because the composites now have more structural variability. The improved textural features promote dye adsorption and

Table I Brunauer–Emmett–Teller (BET) surface area, total pore volume, and average pore diameter of CZFO and CZFO–CN composites

Sample	BET surface area, m^2/g	Total pore volume, cm^3/g	Average pore diameter, nm
CZFO	22.6	0.147	15.8
CZFO-CN15	28.4	0.182	17.2
CZFO-CN25	33.7	0.213	18.6

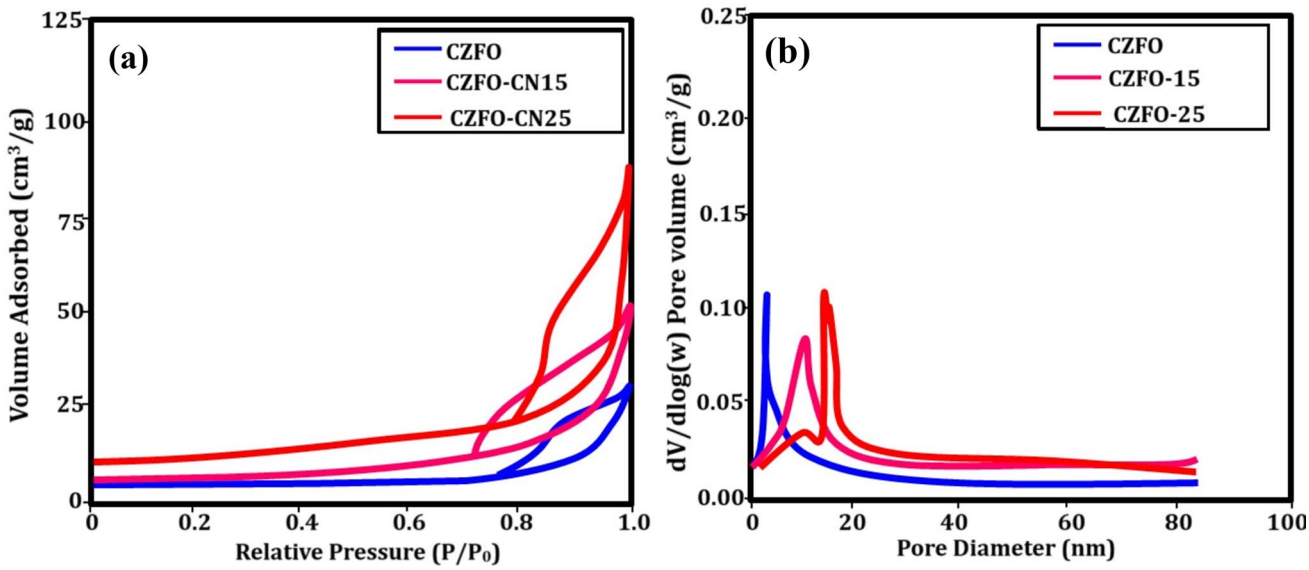


Fig. 5 (a) N_2 adsorption–desorption isotherms (b) Barrett–Joyner–Halenda (BJH) pore size distribution curves of CZFO, CZFO–CN15 and CZFO–CN25.

reactant diffusion in photocatalysis, while also markedly elevating the photovoltaic performance of the CZFO–CN composites. The interconnected mesoporous structure facilitates effective electrolyte penetration and ion movement, thus diminishing charge-transfer resistance and enhancing catalytic performance at the counter-electrode–electrolyte interface. The synergistic effects immediately enhance the power-conversion efficiency of CZFO–CN-based DSSCs.[41]

UV and PL Analysis

The optical absorption properties of CZFO, CZFO–CN15, CZFO 20, and CZFO–CN25 were studied using UV and PL analysis. Figure 6a presents the UV–Vis diffuse reflectance spectra of CZFO, CZFO–CN15, and CZFO–CN25. Pristine CZFO has an absorption edge of about 580 nm, indicating a bandgap of roughly 2.57 eV. The observed red-shift of the absorption edge from ≈580 nm for pristine CZFO (E_g≈2.57 eV) to ≈620 nm for CZFO–CN25 (E_g≈2.20 eV) indicates effective interfacial bandgap engineering. The hybridization between CZFO and g-C_3N_4 introduces localized electronic states near the Fermi level, narrowing the apparent bandgap and extending visible-light absorption. This modulation originates from orbital overlap and surface polarization at the heterojunction rather than bulk doping, consistent with the unchanged XRD peaks. Moreover, the calculated absorption coefficients (α) for CZFO, CZFO–CN15, CZFO–CN20, and CZFO–CN25 were approximately 3.8×10^4 cm^{-1}, 4.5×10^4 cm^{-1}, 5.2×10^4 cm^{-1}, and 5.8×10^4 cm^{-1}, respectively. The steady increase in α values with higher g-C_3N_4 content confirms the enhanced light-harvesting capability of the composites, which supports the observed improvement in photocatalytic and photovoltaic performance.[42] The inclusion of g-C_3N_4 causes the absorption margins of CZFO–CN15 and CZFO–CN25 to move towards longer wavelengths, signifying enhanced visible light absorption. The calculated optical bandgaps were diminished to 2.41 eV, 2.32 eV, and 2.20 eV for CZFO–CN15, CFZO-CN20, and CZFO–CN25, respectively, according to Tauc plot extrapolation (Fig. 6b).[43] The decrease in bandgap results from robust interfacial electronic interactions between CZFO and g-C_3N_4, facilitating charge redistribution and the emergence of localized mid-gap states at the heterojunction interface. The newly formed electronic states diminish the energy necessary for photoexcitation, thereby effectively constricting the optical bandgap. As a result, the heterostructure demonstrates augmented light absorption and enhanced charge-carrier separation, advantageous for photocatalytic and photovoltaic applications under visible-light exposure.[44] The photoluminescence (PL) spectra of g-C_3N_4, CZFO, and CZFO–CN composites were obtained at an excitation wavelength of 325 nm, aligning with the band-to-band excitation energy of both CZFO and g-C_3N_4 (Fig. 6c). This wavelength was chosen as it occurs within the robust UV absorption spectrum of g-C_3N_4 (≈300–380 nm), facilitating effective excitation of photoinduced charge carriers and allowing for the evaluation of recombination dynamics in the hybrid systems. As shown in the figure, all the samples show broad emission bands in the visible range. These bands typically occur by photogenerated charge carriers recombining. The strongest emission is seen in pure CZFO, which means that electron–hole pairs are recombining quickly. When g-C_3N_4 is added to the CZFO–CN15, CFZO-CN20, and CZFO–CN25 mixtures, the emission level goes down significantly. In particular, CZFO–CN25 has the weakest PL strength, which means it stops recombination the best. The PL quenching observed for CZFO–CN25 further verifies that this interfacial band alignment promotes efficient charge separation and suppresses electron–hole recombination.[45] The efficient separating of charges allows for breaking down dyes photocatalytically and move charges around in DSSC devices, which improves their overall performance.

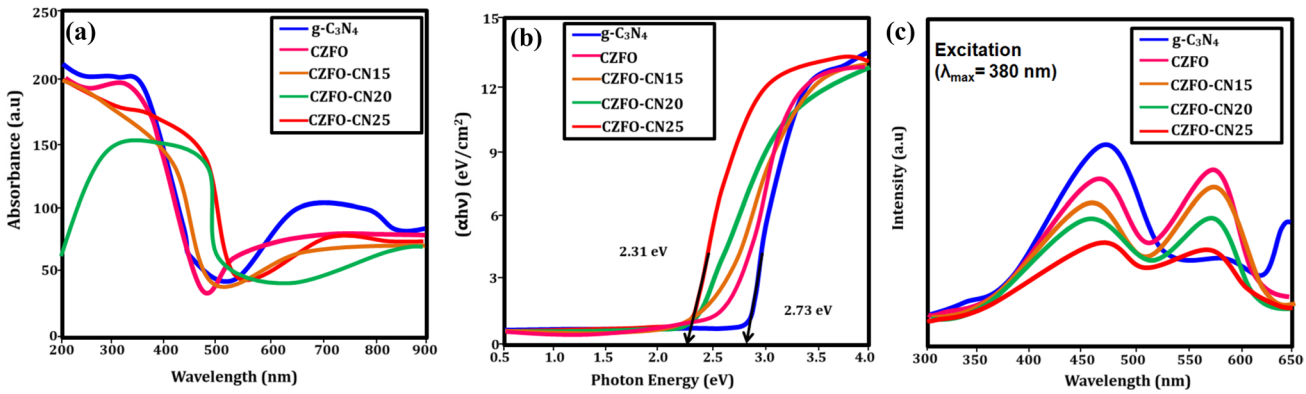


Fig. 6 (a) UV–Vis diffuse reflectance spectra (DRS) spectra (b) Tauc plots for bandgap estimation (c) PL spectra of g-C_3N_4, CZFO, CZFO–CN15, and CZFO–CN25.

Photocatalytic Activity Analysis

The photocatalytic degradation performance of the synthesized samples was evaluated using RhB as the model pollutant under natural sunlight irradiation. RhB was selected as the model dye pollutant because to its prominent chromophoric absorption band, exceptional stability, and common application as a standard reference for evaluating visible-light photocatalysts.[46] This selection facilitates a direct comparison of the photocatalytic efficiency and charge-transfer performance of the newly synthesized CZFO–CN composites with previously documented systems. The principal objective of this work was to assess the structure–activity connection and interfacial charge-separation dynamics, rather than to attain instantaneous drinkable water quality. To validate the reproducibility of the degradation investigation, spectral monitoring of RhB decolorization and total organic carbon (TOC) measurements were conducted, confirming significant mineralization of organic residues during the photocatalytic process. These evaluations confirm that the detected dye loss results from authentic photocatalytic oxidation, rather than simple adsorption or color degradation. Prior to evaluating the composites, control experiments were conducted using pristine CZFO and pristine g-C_3N_4 under identical experimental conditions (50 mg catalyst, 100 mL of 10 mg L^1 RhB solution, natural sunlight, pH≈6.5). As shown in Fig. 7a, pristine CZFO achieved a degradation efficiency of 61.4%, while pure g-C_3N_4 exhibited only 38.5% removal after 90 min of irradiation. These low efficiencies confirm that neither component alone can achieve significant degradation. In contrast, the CZFO–CN composites exhibited markedly enhanced activity, demonstrating the essential role of interfacial coupling and charge transfer between CZFO and g-C_3N_4 in driving the photocatalytic process. Among the samples, the CZFO–CN25 composite exhibited the highest degradation efficiency of 94.6% within 90 min, while CZFO–CN15 and pristine CZFO achieved 78.2% and 61.4% degradation, respectively. The superior photocatalytic performance of the composites is mainly attributed to their enhanced UV–Visible absorption, suppressed electron–hole recombination (PL), and increased surface area, which synergistically improve light harvesting and reactive site availability. To further understand the degradation behavior, the kinetic data were fitted using the pseudo-first-order model ln $(C_0/Ct) = kt$, as illustrated in Fig. 7b.[47] The calculated rate constants were 0.0123 min^{-1} (CZFO), 0.0195 min^{-1} (CZFO–CN15), 0.0255 min^{-1} (CZFO–CN20) and 0.0276 min^{-1} (CZFO–CN25), indicating a substantial enhancement in reaction kinetics with increasing g-C_3N_4 content. The reusability and stability of the CZFO–CN25 photocatalyst were assessed over five successive degradation cycles. As shown in Fig. 7c, the material retained over 90% of its original activity, demonstrating excellent stability and recyclability under sunlight. Post-SEM analyses confirmed the structural integrity of the composite after cycling, reinforcing its potential for long-term use in practical applications (Fig. 7d. Furthermore, optimization studies were also conducted to identify ideal

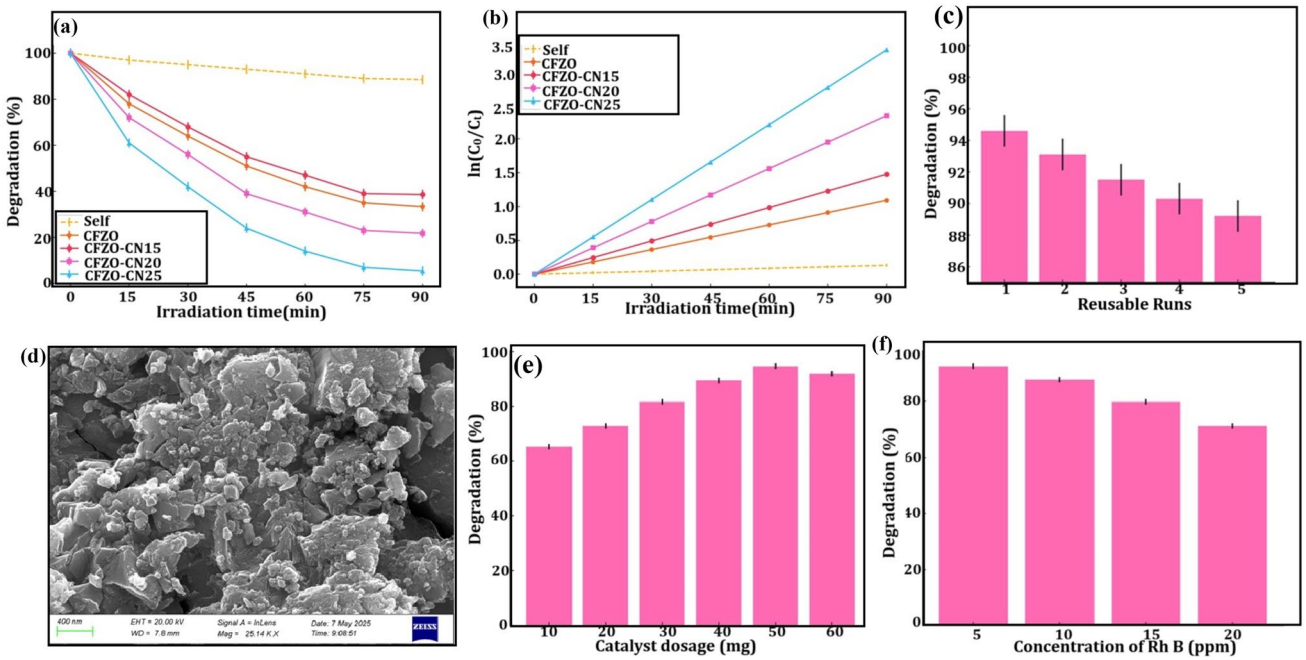


Fig. 7 (a) Degradation of RhB under sunlight ;(b) kinetic plot ;(c) reusability over five cycles ;(d) SEM image of CZFO–CN25 after five cycles; (e) effect of catalyst dosage (f) effect of dye concentration.

operational conditions. As shown in Fig. 7e, the effect of catalyst dosage was studied by varying CZFO–CN25 from 10 to 60 mg per 100 mL of dye solution. The degradation efficiency increased with dosage, peaking at 50 mg, beyond which it slightly decreased owing to light scattering and reduced penetration.[48] Similarly, Fig. 7f shows the effect of initial dye concentration, where lower RhB concentrations (5–10 mg/L) resulted in higher degradation owing to better light absorption and availability of active sites. At higher concentrations (15–20 mg/L), the efficiency declined due to surface saturation and limited light activation.[49] These findings highlight that the CZFO–CN25 composite exhibits excellent photocatalytic performance, optimal activity at 50 mg catalyst loading, and high durability making it a promising photocatalyst for visible-light-driven degradation of organic pollutants.

Photovoltaic Analysis

The J–V characteristics of DSSCs fabricated using Pt, CZFO, CZFO–CN15, and CZFO–CN25 CE were measured under simulated AM 1.5G sunlight with an intensity of 100 mW cm^{-2}. The resulting curves are shown in Fig. 8a and the corresponding photovoltaic parameters such as short-circuit current density (J_{sc}), open-circuit voltage (V_{oc}), fill factor (FF) and PCE are summarized in Table II. Among the samples, the DSSC with the CZFO–CN25 CE exhibited the best performance, achieving a PCE of 6.14%, with a J_{sc} of 15.06 mA cm^{-2}, V_{oc} of 0.75 V, and FF of 0.64. In contrast, CZFO–CN15 and CZFO devices showed relatively lower efficiencies of 5.13% and 3.84%, respectively. The reference Pt-based device reached a slightly higher efficiency of 6.56%, as expected from its superior catalytic activity.

Table II Photovoltaic performance parameters of fabricated DSSCs using various counter electrodes

Counter electrode	J_{sc}, mA/cm^2	V_{oc}, V	Fill factor	PCE, %
CZFO	10.02	0.64	0.65	3.84
CZFO-CN15	12.09	0.68	0.63	5.13
CZFO-CN20	14.13	0.72	0.58	5.74
CZFO-CN25	15.06	0.75	0.64	6.14
Pt	15.98	0.78	0.64	6.56

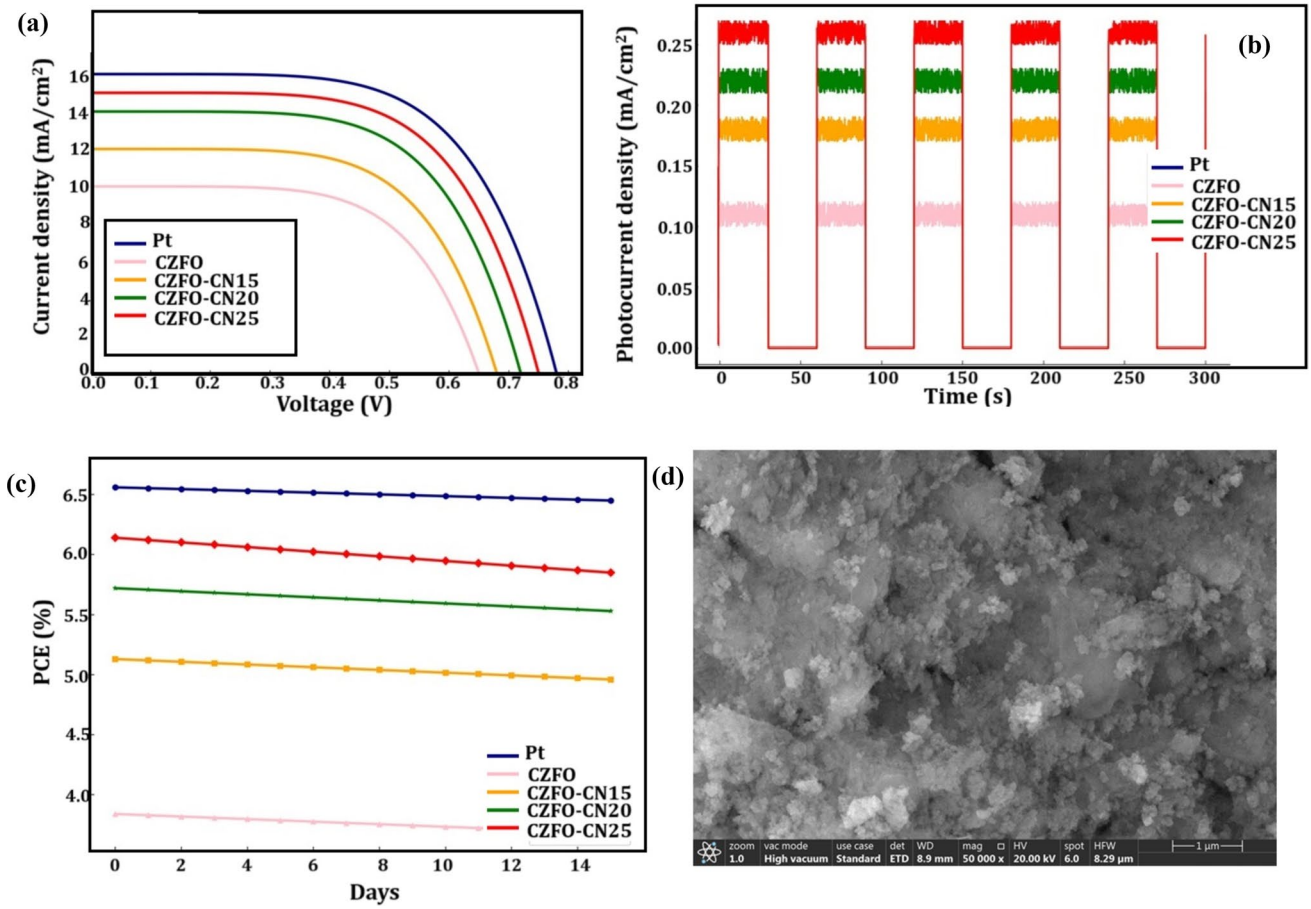


Fig. 8 (a) J–V curves of DSSCs; (b) Transient photocurrent response; (c) stability of CZFO–CN25 CE; (d) SEM image after stability testing.

The enhanced photovoltaic efficacy of the CZFO–CN composite-based DSSCs is primarily owing to the synergistic interaction between CZFO and g-C_3N_4.[50] The integration of g-C_3N_4 not only boosts the composite's electrical conductivity but also provides extra catalytic active sites, thereby facilitating the effective reduction of triiodide. Furthermore, the hybridization facilitates improved electron transfer at the interface between the CE and the electrolyte, thereby reducing charge recombination. The composites' mesoporous nature, which is further substantiated by BET analysis, enables the electrolyte to penetrate more effectively and expedites ion transport.[51] The CZFO–CN25-based DSSC demonstrated the highest efficiency among all the tested electrodes, virtually approaching that of the Pt-based cell, while providing a more economical and stable alternative. The transient photocurrent response of the fabricated CE was recorded under visible light irradiation using a standard three-electrode setup in 0.1 M Na_2SO_4 electrolyte, with the CZFO, CZFO–CN15, and CZFO–CN25 electrodes acting as working electrodes. A Pt wire and Ag/AgCl were used as the counter and reference electrodes, respectively. As shown in Fig. 8b, all samples demonstrated reproducible ON–OFF cycles, indicating a stable and responsive photocurrent behavior upon light exposure. Among the tested electrodes, CZFO–CN25 exhibited the highest photocurrent density of approximately 0.26 mA cm^{-2}, which was significantly higher than CZFO–CN15 (0.18 mA cm^{-2}) and CZFO (0.11 mA cm^{-2}). This enhancement is attributed to the effective charge separation and improved charge transport at the CZFO/g-C_3N_4 heterojunction interface, along with increased active surface area confirmed by BET analysis.[52] To examine the operational durability of the fabricated CE, a stability test was carried out for the CZFO–CN25-based device over a 15-day period under ambient conditions. As shown in Fig. 8c, the device retained approximately 95.2% of its initial PCE after storage, indicating excellent temporal stability. The minimal drop in performance suggests that the electrode maintains its structural and electrocatalytic integrity even after extended exposure. Furthermore, SEM analysis was also performed to look for any morphological changes after prolonged use. As shown in Fig. 8d, without noticeable aggregation, surface cracking and deterioration, the CZFO–CN25 surface preserved its nanoscale design. This suggests that even after extensive cycling and ageing, the hybrid structure maintains its structural stability. These findings demonstrate the CZFO–CN25 counter electrode's mechanical stability and its potential as a long-term, reasonably priced alternative to traditional Pt-based electrodes in DSSC applications.

Electrochemical Evaluation of Fabricated CE's

To assess the electrocatalytic activity and interfacial charge transfer characteristics of the fabricated counter electrodes, cyclic voltammetry (CV), Tafel polarization, and electrochemical impedance spectroscopy (EIS) were performed using symmetrical dummy cells in an iodide/triiodide (I^-/I_{3-}) electrolyte system. The redox kinetics of the I^-/I_{3-} couple were analyzed using CV and the results are shown in Fig. 9a. As shown in the figure, the CZFO–CN25 composite exhibited the highest cathodic peak current density of −2.35 mA cm^{-2} and a relatively narrow peak-to-peak separation (ΔE_p) of 390 mV, outperforming all other fabricated electrodes in terms of electrocatalytic activity. These findings suggest that the triiodide reduction process is both reversible and efficient, emphasizing the favorable electron transfer kinetics at the electrode–electrolyte interface.[53] The synergistic effect between CZFO and g-C_3N_4 is responsible for the improved performance of CZFO–CN25. The incorporation of g-C_3N_4 promotes better charge transfer, increases the number of active sites and enhances electrical conductivity.[54] In contrast, the pristine CZFO and CZFO–CN15 composites exhibited weaker electrocatalytic behavior, with cathodic current densities of 1.68 mA cm^{-2} and −2.04 mA cm^{-2}, respectively, and larger ΔE_p values of 429 mV and 422 mV. The slower redox kinetics and limited catalytic efficacy are inferred from the broader separations and lower current responses. As anticipated, the platinum

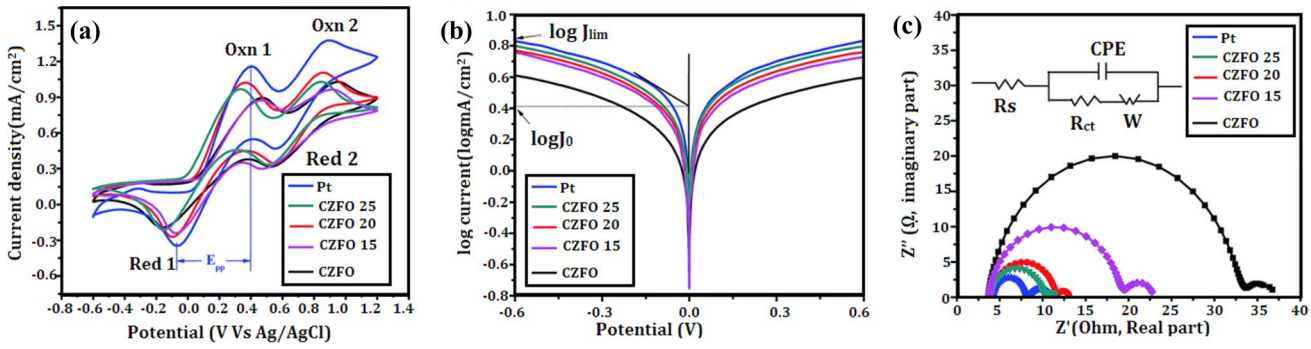


Fig. 9 (a) CV curves (b) Tafel polarization plots (c) Nyquist plots for various counter electrodes.

electrode demonstrated the most promising performance, with a cathodic current density of −2.72 mA cm^{-2} and a ΔE^p of 371 mV, thereby serving as a benchmark for comparison. Nevertheless, the near-comparable activity of CZFO–CN25 is a testament to its potential as a cost-effective and durable alternative to Pt-based electrodes in DSSC applications. Moreover, the charge-transfer kinetics at the electrode–electrolyte interface were assessed using Tafel polarization measurements, and the results are depicted in Fig. 9b. The Tafel slope is derived from the linear segment of the overpotential (η) versus logarithm of current density (log J) plot in these measurements.[55] The exchange current density (J_0), a crucial measure of catalytic activity, is derived from the linear segment by projecting the fitted line to zero overpotential ($\eta = 0$), where log J aligns with log J_0. An elevated J_0 value indicates accelerated redox kinetics and enhanced catalytic efficacy. Among the analyzed samples, the CZFO–CN25 electrode demonstrated a J_0 value of 0.92 mA cm^{-2}, signifying effective electron transport and improved electrocatalytic performance. This value exceeds those of CZFO–CN15 (0.71 mA cm^{-2}) and virgin CZFO (0.53 mA cm^{-2}), indicating the contribution of g-C_3N_4 in enhancing conductivity and offering supplementary active sites.[56] The Pt electrode attained the maximum exchange current density of 1.08 mA cm^{-2}, hence validating its exceptional redox kinetics. The comparatively elevated J_0 found in the CZFO–CN25 composite indicates its competitive efficacy as a cost-effective substitute for Pt in DSSC counter electrodes. In addition, Fig. 9c shows the Nyquist plots used for EIS analysis. These plots help to determine the charge-transfer resistance (R_{ct}) and series resistance (R_s) values. More effective charge transfer at the electrode–electrolyte interface was confirmed by the CZFO–CN25 electrode's much lower R_{ct} of 6.25 Ω, which was lower than the 9.81 Ω for CZFO–CN15 and the 15.36 Ω for CZFO. The Pt electrode's R_{ct} of 5.12 Ω was marginally superior to that of CZFO–CN25. Consistent cell design and excellent component contact were indicated by the series resistance (R_s), which were almost constant across all samples (~4.2–4.8 Ω). A summary of all electrochemical parameters, including cathodic peak current, ΔE_p, J_0, R_{ct}, and R_s, is presented in Table III. These results clearly demonstrate that the CZFO–CN25 CE exhibits electrocatalytic performance comparable to that of Pt, while offering a more cost-effective and stable alternative for DSSC applications. The morphological characteristics of the CZFO–CN25 electrode can be directly correlated with its enhanced electrochemical performance. The FESEM and TEM images demonstrate a porous and interconnected surface, which facilitates rapid electrolyte diffusion and provides a significant number of exposed active sites. The efficient transport of ions and electrons is facilitated by this structural porosity, as evidenced by the lower R_{ct} and a greater exchange current density (J_0) as observed in Tafel and EIS analyses. Consequently, the improved electrochemical properties are a result of the synergistic effect of efficient charge-transfer kinetics and favorable morphology.

Mechanism of Enhanced Photocatalytic and Photovoltaic Performance of CZFO–CN25

The superior performance of the CZFO–CN25 hybrid material in both photocatalytic degradation and DSSC applications is primarily attributed to the interfacial coupling and bandgap engineering between CZFO and g-C_3N_4. The formation of a well-integrated type-II heterojunction promotes efficient charge separation, improved light absorption, and enhanced redox reactivity. In the photocatalytic process, both CZFO and g-C_3N_4 absorb visible photons and generate electron–hole pairs. Because the conduction band (CB) edge of g-C_3N_4 (≈ −1.13 eV) is more negative than that of CZFO (≈ −0.53 eV), photoexcited electrons in g-C_3N_4 readily migrate to the CB of CZFO, while the corresponding holes remain in the valence band (VB) of g-C_3N_4 (≈ +1.57 eV).[57–59] This directional charge migration effectively suppresses recombination, consistent with the significant PL quenching observed for the composite. The transferred electrons on CZFO reduce adsorbed O_2 to generate superoxide radicals (•O_{2-}), whereas holes on g-C_3N_4 oxidize water or hydroxyl groups to form hydroxyl radicals (•OH). These reactive oxygen species cooperatively degrade RhB into smaller, nontoxic intermediates. The improved absorption (red-shift from 580 nm to 620 nm) and narrowed bandgap (2.57 → 2.20 eV) further confirm the interfacial bandgap modulation that enhances visible-light harvesting

Table 3 Electrochemical parameters from CV, Tafel, and EIS analyses

CE	Cathodic peak current, mA/cm^2	E_{pp}, mV	Exchange current density (J_0), mA/cm^2	Limiting current density (J_{lim}), mA/cm^2	R_{ct}, Ω/cm^2	R_s, Ω/cm^2
Pt	−2.72	371	1.08	2.73	5.12	4.2
CZFO	−1.68	429	0.53	1.64	15.36	4.8
CZFO-CN15	−2.04	422	0.71	1.90	9.81	4.6
CZFO-CN20	−2.21	417	0.82	2.07	8.32	4.5
CZFO-CN25	−2.35	390	0.92	2.32	6.25	4.3

and accelerates the photocatalytic reaction. In the photovoltaic process, the same heterojunction synergy facilitates efficient electron transport and redox kinetics at the counter-electrode–electrolyte interface. The well-anchored CZFO nanoparticles on the conductive g-C_3N_4 matrix provide abundant catalytic sites and conductive pathways, reducing charge-transfer resistance ($R_{ct} = 6.25\ \Omega$) and increasing exchange current density ($J_0 = 0.92$ mA cm^{-2}). This leads to superior electrocatalytic performance for triiodide reduction and a higher power conversion efficiency (6.14%), nearly matching that of the Pt-based cell.

Overall, the synergistic interfacial coupling, bandgap realignment, and efficient charge separation between CZFO and g-C_3N_4 are the key factors responsible for the dual enhancement in photocatalytic and photovoltaic performance. The proposed mechanism is illustrated schematically in Fig. 10a and b, showing the directional transfer of electrons from the CB of g-C_3N_4 to that of CZFO and hole retention in the VB of g-C_3N_4, enabling prolonged carrier lifetimes and enhanced reactivity. In the photovoltaic application, the CZFO–CN composite acts as a CE facilitating the reduction of I_{3-} to I^- in the electrolyte. The integration of g-C_3N_4 improves the electrical conductivity and catalytic activity by offering additional active sites and enhancing charge transport at the electrode–electrolyte interface.[60] This is supported by CV results showing increased peak currents and smaller ΔE_p values for CZFO–CN25, along with Tafel plots indicating a higher exchange current density (J_0). EIS analysis also confirms the lowest charge-transfer resistance (R_{ct}) for CZFO–CN25, highlighting efficient interfacial electron transfer. Furthermore, BET results reveal increased surface area and porosity in the composite, which promote better dye/electrolyte interaction in DSSCs and more reactive sites in photocatalysis.[61] The consistent light response in I–t curves and stability over time reaffirm the structural robustness and electron transport capability of the composite. Overall, the formation of an efficient heterojunction, improved conductivity, enhanced surface area, and robust charge dynamics collectively contribute to the dual-functionality of CZFO–CN25, making it a promising material for environmental remediation and sustainable energy conversion.

Comparative Analysis with Previously Reported Materials

The photovoltaic and photocatalytic performance of the CFZO–CN25 composite was benchmarked against a variety of reported dual-function materials in order to comprehensively assess its multifunctional efficacy, as summarized in Table IV.[62–71] CFZO–CN25 demonstrated the highest PCE of 6.14% among all the materials enumerated, surpassing extensively researched systems such as Cu–TiO_2 (3.9%), MoS_2–NiO–CuO (4.92%), and even Cu–Ni–Titania (5.6%). The heterojunction structure of CZFO and g-C_3N_4 enables enhanced interfacial charge transfer and excellent electrocatalytic activity, as evidenced by the superior photovoltaic performance. This is further substantiated by CV, EIS, and Tafel analyses. CFZO–CN25 exhibited a degradation efficiency of 94.6% for RhB in terms of photocatalytic capability, placing it among the top-performing composites. This performance is comparable to that of CdTe–MOF (95%), Cu–Ni–Titania (99%) and $FeCo_2O_4$ (99%). Nevertheless, the CFZO–CN25 composite is a safer and more sustainable alternative than Cd- or heavy metal-based systems owing to its complete composition of earth-abundant and environmentally benign elements. It is important to note that a significant number

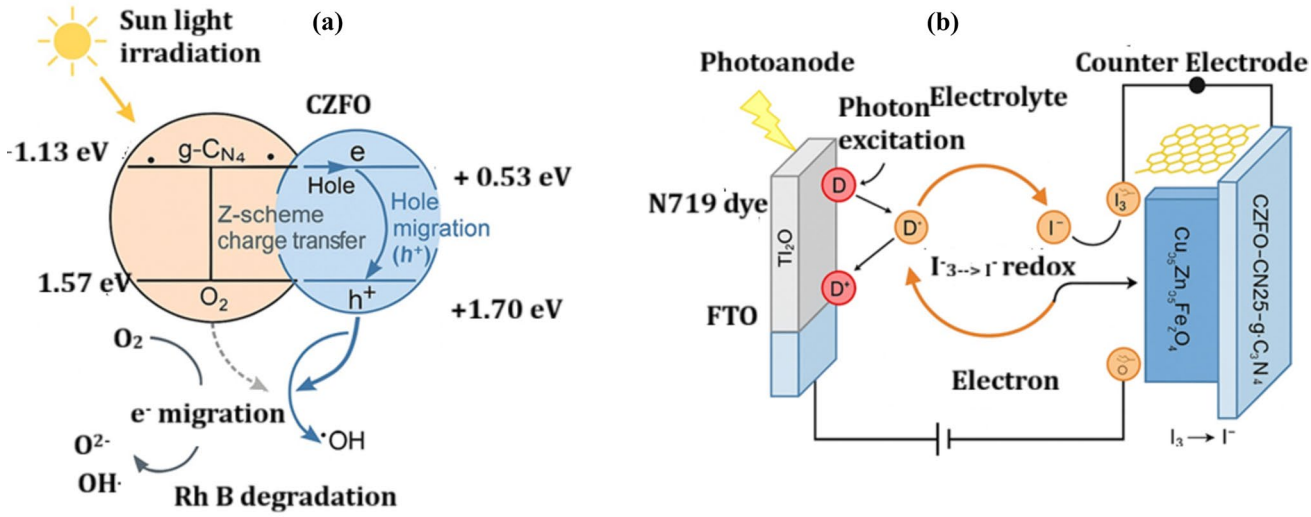


Fig. 10 Schematic illustration of the proposed mechanism for (a) photocatalytic degradation mechanism (b) photovoltaic mechanism.

Table IV Comparative performance of CZFO–CN25 with previously reported materials for photocatalytic and DSSC applications

Material	Application	PCE%/degradation%	Reference
Cu–TiO_2	DSSC and photocatalyst	PCE 3.9% and 97.2% for RhB removal	62
$SrTiO_3$	DSSC and photocatalyst	PCE of 1.12% and 36% for MB removal	63
MoS_2/NiO/CuO	DSSC and photocatalyst	PCE of 4.92% and 93% of MO removal	64
MoS_2 Nanosheet	DSSC and photocatalyst	PCE of 3.81% and 91.84% of MB removal	65
CdTe/MOF composite	DSSC and photocatalyst	PCE of 3.20% and 95% of Rh 6G removal	66
$FeCo_2O_4$	DSSC and photocatalyst	PCE of 3.31% and 99% of MB removal	67
Cu–Ni–Titania	DSSC and photocatalyst	PCE of 5.6% and 99% of Rh B removal	68
Fe^{3+}–$SrTiO_3$	DSSc and photocatalyst	PCE of 3.23% and 82% of MB removal	69
CdSe–ZnO–TiO_2–CuO	DSSc and photocatalyst	PCE of 3.6% and 96% of MB removal	70
Co-Mn co doped TiO_2	DSSc and photocatalyst	PCE of 4.99% and 87.8% of MB removal	71
CFZO–CN25	DSSC/photocatalyst	6.14%/94.6%	This work

of the reported materials have either a lower PCE (e.g., $SrTiO_3$ with 1.12%, Fe^{3+}–$SrTiO_3$ with 3.23%) or moderate degradation efficiency (e.g., Co-Mn-TiO_2 with 87.8%), which restricts their multifunctional performance in practical applications. The potential of CFZO–CN25 as a cost-effective, dual-functional material is demonstratively demonstrated by its ability to combine high photovoltaic efficiency with strong photocatalytic degradation under visible light, in addition to its proved structural and operational stability. In response to the increasing demand for integrated energy–environmental materials, the current study not only introduces a high-performing Pt-free DSSC counter electrode but also contributes a visible-light-active photocatalyst for environmental remediation.

Conclusions

In summary, the successful synthesis of $Cu_{0.5}Zn_{0.5}Fe_2O_4$–g-C_3N_4 hybrid nanocomposites was achieved through an ultrasonic-assisted hydrothermal route, which resulted in a well-integrated Z-scheme heterojunction. As a Pt-free DSSC counter electrode, the optimized CZFO–CN25 demonstrated an impressive photocatalytic degradation of rhodamine B (94.6% within 90 min) and a power conversion efficiency of 6.14%. Photoluminescence quenching and electrochemical impedance experiments have demonstrated that the synergistic interaction between $Cu_{0.5}Zn_{0.5}Fe_2O_4$ and g-C_3N_4 facilitated efficient charge separation, reduced recombination, and enhanced redox reactivity. These findings confirm that the key to achieving multifunctional performance is rational heterojunction engineering and thin-film optimization. The current study not only illustrates a sustainable alternative to noble-metal electrodes but also establishes a design framework for the integration of photocatalytic and photovoltaic functionalities in a singular hybrid material. Future endeavors may concentrate on the extension of this methodology to other ferrite–carbon nitride systems and the enhancement of interfacial architectures to facilitate the fabrication of large-area, flexible devices.

Acknowledgements The author(s) received no financial support for the research, authorship, and/or publication of this article.

Author Contributions V.A., performed conceptualization, methodology, writing—review and editing. S.T., performed formal analysis and data curation. M.M.A., performed data validation. M.H.K., carried out original draft writing and supervision of this project.

Funding No funding obtained for this study.

Data Availability All data and information generated during this study are available and will be made upon reasonable request.

Conflict of Interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

1. A. Garg, and L. Chopra, Dye waste: a significant environmental hazard. *Mater. Today Proc.* 48, 1310–1315 (2022).
2. H. Ali, Biodegradation of synthetic dyes—A review. *Water Air Soil Pollut.* 213, 251–273 (2010).
3. G. Thennarasu, S. Rajendran, A. Kalairaj, H.S. Rathore, R.C. Panda, and T. Senthilvelan, A comprehensive review on the application of semiconductor nanometal oxides photocatalyst for the treatment of wastewater. *Clean Technol. Environ. Policy* 27(2), 495–516 (2025).
4. A.S. Morshedy, E.M. El-Fawal, T. Zaki, A.A. El-Zahhar, M.M. Alghamdi, and A.M. El Naggar, A review on heterogeneous photocatalytic materials: Mechanism, perspectives, and environmental and energy sustainability applications. *Inorg. Chem. Commun.* 163, 112307 (2024).
5. B. O'regan, and M. Grätzel, A low-cost, high-efficiency solar cell based on dye-sensitized colloidal TiO_2 films. *Nature* 353(6346), 737–740 (1991).
6. M. Grätzel, Dye-sensitized solar cells. *J. Photochem. Photobiol., C* 4(2), 145–153 (2003).
7. M. Dayanithy, G.C. Sekhar, R.C. Thivyarathi, P. Divyabharathi, L.P. Maguluri, V. Saravanan, Synergistic integration of $NiCo_2O_4$

and $NiWO_4$ nanosheets on Ni foam for advanced supercapacitor applications. *J. Porous Mater.* 1-11 (2025).
8. K. Malaie, Z. Heydari, and M.R. Ganjali, Spinel nano-ferrites as low-cost (photo) electrocatalysts with unique electronic properties in solar energy conversion systems. *Int. J. Hydrogen Energy* 46(5), 3510–3529 (2021).
9. R. Valenzuela, Novel applications of ferrites. *Phys. Res. Int.* 2012(1), 591839 (2012).
10. Y. Liu, J. Hsu, Y. Fu, and K. Tsai, Preparation of Cu–Zn ferrite photocatalyst and it's application. *Int. J. Hydrogen Energy* 41(35), 15696–15702 (2016).
11. B.A.Y.R.A.M. Ünal, M.A. Almessiere, I.A. Auwal, Y. Slimani, and A. Baykal, The role of Pr3+, Pr3+-Y3+ and Pr3+-Y3+-Dy3+ ions substitutions on the electrical and dielectric properties of NiCo nanospinel ferrites. *J. Alloy. Compd.* 944, 169096 (2023).
12. V. Vignesh, K. Subramani, M. Sathish, and R. Navamathavan, Electrochemical investigation of manganese ferrites prepared via a facile synthesis route for supercapacitor applications. *Colloids Surf. A* 538, 668–677 (2018).
13. R. Acharya, S. Pati, and K. Parida, A review on visible light driven spinel ferrite-g-C_3N_4 photocatalytic systems with enhanced solar light utilization. *J. Mol. Liq.* 357, 119105 (2022).
14. J. Dong, Y. Zhang, M.I. Hussain, W. Zhou, Y. Chen, and L.N. Wang, g-C_3N_4: properties, pore modifications, and photocatalytic applications. *Nanomaterials* 12(1), 121 (2021).
15. J. Safaei, N.A. Mohamed, M.F.M. Noh, M.F. Soh, N.A. Ludin, M.A. Ibrahim, and M.A.M. Teridi, Graphitic carbon nitride (gC_3N_4) electrodes for energy conversion and storage: a review on photoelectrochemical water splitting, solar cells and supercapacitors. *J. Mater. Chem. A* 6(45), 22346–22380 (2018).
16. M. Mahmood, M.A. Yousuf, M.M. Baig, M. Imran, M. Suleman, M. Shahid, and M.F. Warsi, Spinel ferrite magnetic nanostructures at the surface of graphene sheets for visible light photocatalysis applications. *Phys. B Condens. Matter.* 550, 317–323 (2018).
17. T. Aravind, A. Rajasekaran, R. Arun, G. Elumalai, and M. Bindhu, Strategic design of 2D graphitic carbon nitride nanosheets anchored with $CuFe_2O_4$ nanoparticles for efficient photoanodes in DSSC applications. *Diam. Relat. Mater.* 154, 112091 (2025).
18. M. Thangamuthu, Q. Ruan, P.O. Ohemeng, B. Luo, D. Jing, R. Godin, and J. Tang, Polymer photoelectrodes for solar fuel production: progress and challenges. *Chem. Rev.* 122(13), 11778–11829 (2022).
19. S. Huang, Q. Dong, Y. Lu, L. Duan, and D. Zhang, Outstanding performance of electron-transport-layer-free perovskite solar cells using a novel small-molecule interlayer modified FTO substrate. *Chem. Eng. J.* 422, 130001 (2021).
20. L. Sun, T. Zhang, Y. Liu, J. Wang, W. Zong, Y. Cao, and X. Wang, Hierarchical NiSe microspheres as high-efficiency counter electrode catalysts for triiodide reduction reaction. *Sol. Energy* 225, 486–493 (2021).
21. Y. Lv, C. Gong, Y. Dong, and H.J. Choi, Synthesis of rGO/$CoFe_2O_4$ composite and its magnetorheological characteristics. *Materials* 17(8), 1859 (2024).
22. Y. Zhang, Q. Pan, G. Chai, M. Liang, G. Dong, Q. Zhang, and J. Qiu, Synthesis and luminescence mechanism of multicolor-emitting g-C_3N_4 nanopowders by low temperature thermal condensation of melamine. *Sci. Rep.* 3(1), 1943 (2013).
23. Y. Zhang, A.K. Selva Kumar, D. Li, M. Yang, and R.G. Compton, Nanoparticle-and nanotube-modified electrodes: response of drop-cast surfaces. *ChemElectroChem* 7(22), 4614–4624 (2020).
24. P. Sivakarthik, V. Thangaraj, K. Perumalraj, and J. Balaji, Synthesis of co-doped tin oxide nanoparticles for photo catalytic degradation of synthetic organic dyes. *Dig. J. Nanomater. Biostruct.* 11, 935–943 (2016).
25. Z. Lin, S. Hao, Z. Li, J. Hu, and A. Song, Elucidation of the optical, electronic, and photoelectrochemical properties of *p*-type $CuFe_2O_4$ photocathodes. *Int. J. Hydrogen Energy* 77, 511–519 (2024).
26. T. Xiong, W. Cen, Y. Zhang, and F. Dong, Bridging the g-C_3N_4 interlayers for enhanced photocatalysis. *ACS Catal.* 6(4), 2462–2472 (2016).
27. W.J. Ong, L.L. Tan, Y.H. Ng, S.T. Yong, and S.P. Chai, Graphitic carbon nitride (g-C_3N_4)-based photocatalysts for artificial photosynthesis and environmental remediation: are we a step closer to achieving sustainability? *Chem. Rev.* 116(12), 7159–7329 (2016).
28. G. Sun, Q. Gao, S. Tang, R. Ling, Y. Cai, C. Yu, and A. Wang, Fabrication and enhanced photocatalytic activity of p–n heterojunction $CoWO_4$/g-C_3N_4 photocatalysts for methylene blue degradation. *J. Electron. Mater.* 51(6), 3205–3215 (2022).
29. K.P. Raj, V. Thangaraj, A.P. Uthirakumar, and P. Sivakarthik, Synthesis, chracterization and photocatalytic study of nickel oxide nanoparticles. *Int. J. Sci. Technol.* 2(10), 113 (2014).
30. F. Chang, Y. Xie, C. Li, J. Chen, J. Luo, X. Hu, and J. Shen, A facile modification of g-C_3N_4 with enhanced photocatalytic activity for degradation of methylene blue. *Appl. Surf. Sci.* 280, 967–974 (2013).
31. P.H. Krishnan, R.D. Sul, M.S. Murugan, and V. Saravanan, Optimization of La and Ce mixed metal oxide hybrid for enhanced photoresponse in UV–Visible photosensing applications. *J. Mater. Sci. Mater. Electron.* 36(22), 1377 (2025).
32. S. Anandan, T. Selvamani, G.G. Prasad, A.M. Asiri, and J.J. Wu, Magnetic and catalytic properties of inverse spinel $CuFe_2O_4$ nanoparticles. *J. Magn. Magn. Mater.* 432, 437–443 (2017).
33. K. Baruah, A. Kant, P. Gaijon, S. Ghosh, and M.R. Singh, Synthesis of Cu@ $CuFe_2O_4$ nanocomposite and its application as an efficient catalyst for the production of oxygen. *Appl. Surf. Sci. Adv.* 6, 100130 (2021).
34. J. Zhang, J.M. Song, H.L. Niu, C.J. Mao, S.Y. Zhang, and Y.H. Shen, $ZnFe_2O_4$ nanoparticles: synthesis, characterization, and enhanced gas sensing property for acetone. *Sens. Actuators, B Chem.* 221, 55–62 (2015).
35. V. Saravanan, M. Arthy, S.A. Jayakar, S. Suganya, T. Archana, M. Thirumalai, Hierarchical $ZnCo_2O_4$/MoS_2@ rGO nanocomposite on nickel foam: a promising electrode for advanced energy storage. *Ionics*, 1-13 (2025).
36. Y. Li, S. Wu, L. Huang, J. Wang, H. Xu, and H. Li, Synthesis of carbon-doped g-C_3N_4 composites with enhanced visible-light photocatalytic activity. *Mater. Lett.* 137, 281–284 (2014).
37. X. Yang, F. Qian, G. Zou, M. Li, J. Lu, Y. Li, and M. Bao, Facile fabrication of acidified g-C_3N_4/g-C_3N_4 hybrids with enhanced photocatalysis performance under visible light irradiation. *Appl. Catal. B* 193, 22–35 (2016).
38. V. Arunprasad, P. Rapur, D. Hemanand, M.A. Bennet, and V. Saravanan, Design and fabrication of nanoarchitectures of rGO-decorated $ZnCo_2O_4$ hybrid photocatalyst for high-efficiency solar fuel generation. *J. Mater. Sci. Mater. Electron.* 36(3), 192 (2025).
39. F. Li, L. Xie, G. Sun, F. Su, Q. Kong, Q. Li, and C. Chen, Boosting the specific surface area of hierarchical porous carbon aerogel through the multiple roles of the catalyst for high-performance supercapacitors. *ChemElectroChem* 4(12), 3119–3125 (2017).
40. G. Wu, and B.Q. Xu, Carbon nanotube supported Pt electrodes for methanol oxidation: a comparison between multi-and single-walled carbon nanotubes. *J. Power. Sources* 174(1), 148–158 (2007).
41. D. Sengeni, R. Nadanasabai, V. Saravanan, L. Sangeetha, and J.U. Prakash, Development of binary transition metallic selenide ($NiSe/Co_3Se_4$) hybrid counter electrode for highly efficient Pt-free dye-sensitized solar cells. *Ionics* 30(12), 8281–8293 (2024).
42. N. Kislov, S.S. Srinivasan, Y. Emirov, and E.K. Stefanakos, Optical absorption red and blue shifts in $ZnFe_2O_4$ nanoparticles. *Mater. Sci. Eng. B* 153(1–3), 70–77 (2008).

43. S.S. Nath, K. Chanthirasekaran, S.S. Priya, N. Bharathirja, D. Gupta, S. Kumaran, Boosting electrochemical behavior of biomass-derived porous carbon via N, S, and N, S co-doping for enhanced supercapacitor performance. *Diamond Related Mater.* 112850 (2025).
44. R. Yu, Q. Lin, S.F. Leung, and Z. Fan, Nanomaterials and nanostructures for efficient light absorption and photovoltaics. *Nano Energy* 1(1), 57–72 (2012).
45. F.A. Abuilaiwi, M. Awais, U.Y. Qazi, F. Ali, and A. Afzal, Al3+ doping reduces the electron/hole recombination in photoluminescent copper ferrite ($CuFe_{2-x}Al_xO_4$) nanocrystallites. *Boletín de la Sociedad Española de Cerámica y Vidrio* 61(3), 252–262 (2022).
46. S. Song, B. Cheng, N. Wu, A. Meng, S. Cao, and J. Yu, Structure effect of graphene on the photocatalytic performance of plasmonic $Ag/Ag_2CO_{3-r}GO$ for photocatalytic elimination of pollutants. *Appl. Catal. B* 181, 71–78 (2016).
47. J. Lin, and L. Wang, Comparison between linear and non-linear forms of pseudo-first-order and pseudo-second-order adsorption kinetic models for the removal of methylene blue by activated carbon. *Front. Environ. Sci. Eng. China* 3, 320–324 (2009).
48. K. Prabhu chandran, S.S. Nath, K. Sudha, T. Vinod Kumar, S. Kumaran, J. Sasidevi, Strategic design of TiO_2-SnS hybrid nanostructures: dual-function catalysts for antibiotic removal and solar energy conversion in DSSCs. *J. Electron. Mater.* 1-17 (2025).
49. Y. Zhao, L. Liu, T. Cui, G. Tong, and W. Wu, Enhanced photocatalytic properties of ZnO/reduced graphene oxide sheets (rGO) composites with controllable morphology and composition. *Appl. Surf. Sci.* 412, 58–68 (2017).
50. M. Syamala, D. Srivastava, S.D. Khandekar, T. Porselvi, M.K. Patan, A. Balaram, S. Kumaran, Construction of g-C_3N_4 anchored Cu-ZnS hybrid nanostructures for sustainable energy storage and environmental remediation. *Res. Chem. Intermed.* 1-30 (2025).
51. W. Shu, Y. Liu, Z. Peng, K. Chen, C. Zhang, and W. Chen, Synthesis and photovoltaic performance of reduced graphene oxide–TiO_2 nanoparticles composites by solvothermal method. *J. Alloys Compd.* 563, 229–233 (2013).
52. S. Liang, C. Zhu, N. Zhang, S. Zhang, B. Qiao, H. Liu, and Y. Shi, A novel single-atom electrocatalyst Ti_1/rGO for efficient cathodic reduction in hybrid photovoltaics. *Adv. Mater.* 32(19), 2000478 (2020).
53. N. Dhanasekar, M. Vanitha, N. Shankar, and M. Bindhu, Design and fabrication of $NiSe_2$/g-C_3N_4 hybrid composite counter electrode for Pt-free dye-sensitized solar cells. *J. Mater. Sci. Mater. Electron.* 35(6), 443 (2024).
54. A. Saraswathi, N. Shobanadevi, M. Muthupriya, M.B.M. Yusuf, and T.A. Sheeba, Synergistic incorporation of 2D graphitic carbon nitride into bimetal oxide photoanodes towards higher-performance DSSCs. *J. Electron. Mater.* 53(6), 3384–3397 (2024).
55. S.B. Sundararaj, H. Amir, V. Chinnusamy, and S. Thangavelu, Interfacial charge separation of nickel tungstate anchored on g-C_3N_4 heterojunction stimulates visible-light driven direct Z-scheme photoelectrochemical hydrogen evolution. *Int. J. Hydrogen Energy* 48(67), 26221–26237 (2023).
56. X. Chang, N. Xu, Z. Liu, S. Tian, D. Wen, W. Zheng, and D. Wang, Preparation and performance of g-C_3N_4/CuS film as counter electrode for quantum dot sensitized solar cells. *Process. Appl. Ceramics* 16(2), 167–174 (2022).
57. G. Zhao, J. Ding, F. Zhou, X. Chen, L. Wei, Q. Gao, and Q. Zhao, Construction of a visible-light-driven magnetic dual Z-scheme $BiVO_4$/g-C_3N_4/$NiFe_2O_4$ photocatalyst for effective removal of ofloxacin: mechanisms and degradation pathway. *Chem. Eng. J.* 405, 126704 (2021).
58. G. Gebreslassie, P. Bharali, U. Chandra, A. Sergawie, P.K. Baruah, M.R. Das, and E. Alemayehu, Hydrothermal synthesis of g-C_3N_4/$NiFe_2O_4$ nanocomposite and its enhanced photocatalytic activity. *Appl. Organomet. Chem.* 33(8), e5002 (2019).
59. P. Mishra, A. Behera, D. Kandi, S. Ratha, and K. Parida, Novel magnetic retrievable visible-light-driven ternary Fe_3O_4@$NiFe_2O_4$/phosphorus-doped g-C_3N_4 nanocomposite photocatalyst with significantly enhanced activity through a double-Z-scheme system. *Inorg. Chem.* 59(7), 4255–4272 (2020).
60. T. Zhang, Q. Zhang, Q. Li, F. Li, and L. Xu, Enhanced catalytic $CuCo_2Se_4$/g-C_3N_4 nano-composites as counter electrodes for high-performance quantum dot sensitized solar cells. *Chem. Eng. J.* 454, 140518 (2023).
61. J. Xu, G. Wang, J. Fan, B. Liu, S. Cao, and J. Yu, g-C3N4 modified TiO_2 nanosheets with enhanced photoelectric conversion efficiency in dye-sensitized solar cells. *J. Power. Sources* 274, 77–84 (2015).
62. T. Raguram, and K.S. Rajni, Synthesis and characterisation of Cu-Doped TiO_2 nanoparticles for DSSC and photocatalytic applications. *Int. J. Hydrogen Energy* 47(7), 4674–4689 (2022).
63. K. Aravinthkumar, E. Praveen, A.J.R. Mary, and C.R. Mohan, Investigation on $SrTiO_3$ nanoparticles as a photocatalyst for enhanced photocatalytic activity and photovoltaic applications. *Inorg. Chem. Commun.* 140, 109451 (2022).
64. K. Dharmalingam, A.K. Bojarajan, R. Gopal, E. Thangavel, S.A. Burhan Al Omari, and S. Sangaraju, Direct Z-scheme heterojunction impregnated MoS_2–NiO–CuO nanohybrid for efficient photocatalyst and dye-sensitized solar cell. *Sci. Rep.* 14(1), 14518 (2024).
65. K. Dharamalingam, B.A. Kumar, G. Ramalingam, S.S. Florence, K. Raju, P.S. Kumar, and E. Thangavel, The role of sodium dodecyl sulfate mediated hydrothermal synthesis of MoS_2 nanosheets for photocatalytic dye degradation and dye-sensitized solar cell application. *Chemosphere* 294, 133725 (2022).
66. R. Kaur, A. Rana, R.K. Singh, V.A. Chhabra, K.H. Kim, and A. Deep, Efficient photocatalytic and photovoltaic applications with nanocomposites between CdTe QDs and an NTU-9 MOF. *RSC Adv.* 7(46), 29015–29024 (2017).
67. B. Palanivel, B.A. Kumar, R. Gopal, F.H. Alkallas, M. Shkir, and S. AlFaify, Surfactant assisted $FeCo_2O_4$ nanostructure: an efficient counter electrode for dye sensitized solar cell assisted hexagonal and photocatalyst for dye degradation. *Optik* 280, 170786 (2023).
68. T. Raguram, K.S. Rajni, E. Nandhakumar, and G. Kiruthiga, Investigation on copper–nickel co-doped anatase titania nanospheres as an efficient material for photocatalytic and photovoltaic applications. *New J. Chem.* 47(43), 19963–19973 (2023).
69. K. Aravinthkumar, and C.R. Mohan, Promoting active sites of Fe^{3+} ions in $SrTiO_3$ nanosphere: a superior candidate for high performances of dye-sensitized solar cell and photocatalytic dye degradation. *Colloids Surf. A* 672, 131702 (2023).
70. B. Arjun Kumar, T. Elangovan, D. Karthigaimuthu, D. Aravinth, G. Ramalingam, F. Ran, and S. Sangaraju, CdSe quantum dots bedecked on ZnO/TiO_2/CuO ternary nanocomposite for enhanced photocatalytic and photovoltaic applications. *Langmuir* 39(45), 15864–15877 (2023).
71. A. Ashok, T. Raguram, R.J. Beula, G. Gopinath, S. Ayyasamy, A. Abiram, and B. Vidhya, Synergistic effects of Co-Mn co-doping on the structural and optical properties of TiO_2 nanospheres: dual functions for DSSC photoanodes and degradation photocatalyst. *J. Alloy. Compd.* 1005, 176024 (2024).

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.